Consultancy Report

High Impact Applied Project
MNGMT563-25C
University of Waikato

Executive Summary

Ford Motor Company is facing a material and persistent quality-driven recall and warranty cost crisis. Warranty and field service actions have exceeded one billion dollars in recent periods, with excess warranty costs reaching approximately $1.9 billion in 2023 alone. Regulatory scrutiny has intensified following a $165 million civil penalty related to recall compliance. The issue is not the existence of recalls, which are common in the automotive industry, but the concentration, severity, and recurrence of high-impact events embedded within specific vehicle platforms and component clusters. These patterns indicate structural exposure rather than isolated operational incidents.

Using publicly available NHTSA recall data from 2015 to 2025, this report applied exploratory data analysis and two predictive models under the COSO Enterprise Risk Management framework. A severity-weighted Risk Index was constructed to shift focus from recall frequency to enterprise exposure. The analysis demonstrates that recall risk is heavy-tailed and highly concentrated, with approximately 70–80 percent of severity-weighted exposure driven by a limited number of platform–component clusters. Software, sensing systems, airbags, and transmission-related issues emerged as persistent high-risk subsystems, particularly when embedded within recurring vehicle platforms.

A binary logistic regression model identified platform recurrence and platform–component interaction as the strongest statistical drivers of high-impact recall escalation, with certain platforms exhibiting materially elevated odds of severe outcomes. A Random Forest classification model achieved 95.5 percent accuracy in predicting failed component categories based on recall descriptions, suggesting significant efficiency gains in investigation processes. Together, these findings confirm that Ford’s recall exposure operates as an enterprise governance risk requiring structured prioritization rather than reactive case management.

The recommended course of action is the immediate implementation of a Hotspot-Focused Quality Risk Reduction Program. This strategy concentrates engineering, supplier oversight, and governance resources on the highest-risk platform and component clusters identified in the analysis.

Table of Contents

Problem Definition and Context

"Ford's quality defects are driving safety recalls and warranty costs exceeding over $1B."

Recalls, quality, and warranty costs are closely connected because they represent different stages and severities of the same underlying issue: product defects. Quality problems typically emerge first through manufacturing variation, design weaknesses, supplier defects, or software/hardware integration failures, which then surface in the field as repair incidents and customer complaints. Many of these failures are absorbed through warranty and broader field service actions, where Ford pays to diagnose and repair vehicles during the ownership lifecycle. (Ford Motor Company, 2024) states the following:

> We provide base warranties on the products we sell for specific periods of time and/or mileage, which vary depending upon the type of product and the geographic location of its sale. Separately, we also periodically perform field service actions related to safety recalls, emission recalls, and other product campaigns. Software updates are increasingly a component of vehicle service and may be performed during warranty coverage repairs, through field service actions, or through over-the-air updates. We accrue the estimated cost of both base warranty coverages and field service actions at the time of sale. In addition, from time to time, we issue extended warranties at our expense, the estimated cost of which is accrued at the time of issuance. (p. 87)

When defects are safety-related or noncompliant with regulatory standards, the issue escalates into a recall, making it a formally regulated quality failure requiring remediation at scale. In this sense, recall activity is not separate from quality and warranty outcomes; rather, it represents the subset of quality defects severe enough to trigger regulatory action and coordinated repairs, while warranty and field service actions capture the broader cost footprint of defects that customers experience in normal use. Ford Motor Company is experiencing a sustained quality-and-cost crisis characterized by persistently high safety-recall warranty. Krisher (2024) mention the following regarding Ford's efforts to cut warranty costs:

> In October of 2020, Ford's then-new CEO Jim Farley said the company was working to cut warranty costs after glitch-prone small-car transmissions hit the automaker's bottom line. Nearly four years later, warranty costs are still vexing the nation's second-largest automaker and lopping billions off of its profits.

Muller (2024) comment that “In 2023 alone, Ford logged $1.9 billion in excess warranty costs for vehicles needing repairs”. Expenses went up by 800 million in the second quarter of 2024. Ford Finance Chief John Lawler commented that “most of these warranty expenses were related to older vehicles launched in 2021 or earlier” (Gomes & Eckert, 2024). Also, elevated warranty and field service action costs are materially affecting financial performance and increasing regulatory scrutiny. In November 2024, NHTSA publicly announced a consent order with Ford that includes a $165 million civil penalty and is tied to compliance with federal recall requirements. “The National Highway Traffic Safety Administration announced a consent order with Ford Motor Company for failing to comply with federal recall requirements. The consent order includes a civil penalty of $165 million, the second-largest civil penalty in NHTSA’s history.” (NHTSA, 2024). Regardless of the specific triggering event, a consent order at this scale indicates heightened scrutiny of recall governance and compliance processes. For a global manufacturer, this has implications beyond one campaign: it shapes regulator confidence, compliance costs, and the expectations placed on internal controls and escalation processes. These factors align directly with executive and board oversight responsibilities, because they combine safety outcomes with reputational and financial risk.

The business problem is not simply that recalls occur; recalls are a normal feature of the automotive industry, but that Ford’s frequency, scale, and cost impact indicate systemic quality management deficiencies. (Ford Motor Company, 2025) disclosed an additional large field service action estimate for about $570 million, mentioning the following:

> Ford Motor Company is announcing a field service action related to fuel injectors in certain model year 2021-2024 Bronco Sport vehicles, 2020-2022 Escape vehicles, and 2019-2024 Kuga vehicles. We estimate the aggregate cost of the action, based on the remedy options we are evaluating, to be about $570 million and will be reflected in our second quarter 2025 results.

While Ford characterized the expense as a special item for adjusted reporting purposes, the cash and operational implications remain relevant to enterprise performance. Ford Motor Company (2024) on their 10-k form for the fiscal year ended December 31, 2024, mentions the following regarding warranty cost:

> Ford’s vehicles could be affected by defects that result in recall campaigns, increased warranty costs, or delays in new model launches, and the time it takes to improve the

> quality of our vehicles and services and reduce the costs associated therewith could continue to have an adverse effect on our business. (p. 21)

The issue is operationally consequential because warranty and field service actions consume management attention and resources across engineering, manufacturing, supplier quality, and dealer/service operations. Ford's own reporting explains that warranty and field service action obligations require ongoing estimation and periodic reassessment (Ford Motor Company, 2024) states the following:

> We use historical information regarding the nature, frequency, severity, and average cost of claims for each model year. We assess our obligation for field service actions on a regular basis using actual claims experience and update our estimates as necessary.
>
> Due to the uncertainty and potential volatility of the factors used in establishing our estimates, changes in our assumptions could materially affect our financial condition and results of operations. (p. 87)

This report assumes that publicly available financial disclosures, regulatory actions, and reputable reporting provide sufficient evidence to establish the materiality and urgency of Ford's quality-and-cost challenge, even without access to internal warranty cost allocations by platform or component. The analysis is scoped to a 10-year window (2015–2025) to capture both baseline patterns and recent escalation signals, and it relies primarily on U.S. safety and regulatory datasets as the most transparent public proxy for observable quality outcomes. These assumptions are intended to frame the problem and its significance; causal attribution and prioritization of specific drivers will be addressed through the analytics in subsequent sections.

Methods and Sources

Sources

To discover where Ford's quality problems are, the first step is to find data about the problems. The National Highway Traffic Safety Administration (NHTSA) is a federal agency, part of the U.S. Department of Transportation. The entity oversees setting and enforcing Federal Motor Vehicle Safety Standards (FMVSS) and overseeing vehicle and equipment safety recalls (Focus of the following analysis).

For data collection, the NHTSA website has a collection of Application Programming Interface (API) that contains recall data from all the manufacturers in the United States (National Highway Traffic Safety Administration, n.d.).

The transportation authority has a public dashboard (U.S. Department of Transportation, n.d.), which can be used to filter the recalls based on many variables: manufacturer, component, subject, date of recall, NHTSA ID (Identification of the recall in the Data Base), recall type (If is about a equipment, the vehicle, child seat or tire), potentially affected (Number of vehicles affected), Mfr Campaign Number (manufacturer campaign number – Unique ID of the component in the Manufacturer perspective), recall description, consequence summary (A summary of what might happen), corrective action (What is the action that needs to be taken), park outside advisory (If is advised to park the vehicle outside of a garage), do not drive advisory (If is advised to not drive the vehicle), completion rate % (Percentage of the actual corrections, based on the potentially affected number). For Ford Company, we have recalls since 1966. In this dashboard page, it is possible to export an Excel file (format csv - Comma Separated Values) with all the FORD's recalls.

With this archive, we have the NHTSA ID, the date, Park Outside Advisory, and Do Not Drive Advisory variables that can be used to obtain the severity of the recalls. Based on the NHTSA ID, it is possible to obtain more details of each recall with an Application Programming Interface (API) with a GET method (IBM, n.d.). This method is used to obtain/read more data from another server (In this case, the NHTSA server).

For the second step to obtain the data, the source still is NHTSA, but in this case the data is retrieved through an API request with the GET method (Richardson & Ruby, 2007). Based on the NHTSA recall campaign number, it is possible to recall the details of the recall based on this syntax: api.nhtsa.gov/recalls/campaignNumber?campaignNumber ={campaignNumber}, here is an example of the syntax:

https://api.nhtsa.gov/recalls/campaignNumber?campaignNumber=12V176000

In Appendix A, there is an example of the return call of the syntax.

The result comes in a JSON format and can be transformed into an excel file through a python code (see Appendix B) with the following data: Campaign number, NHTSACampaignNumber, make, model, model year, manufacturer, component, summary, consequence, remedy, notes, report received date, potential number of units affected, parkIt, park outSide, NHTSA Action Number and NHTSA ID. After these two steps, we finally found the data points with 1676 unique campaign numbers and 7498 data points (One campaign number can contain multiple models and each model can contain multiples years of manufacturing).

Justification for the data obtained

The recalls data set is publicly accessible; government data and is available at the NHTSA entity. It does not contain confidential information. Also, the data does not have personal data about the vehicle owners; it is just information about the component failures, manufacturers, dates, etc. This prevents privacy concerns. And last, the data is used for a secondary reason of already collected and de-identified data.

Theoretical Framework: COSO Enterprise Risk Management (ERM)

This report adopts the COSO Enterprise Risk Management (ERM) framework as the primary analytical lens to assess Ford's recall exposure. COSO ERM integrates risk with strategy, performance, governance, monitoring, and reporting (Committee of Sponsoring Organizations of the Treadway Commission [COSO], 2017). Given that Ford's recall activity carries financial, regulatory, and reputational implications, the issue extends beyond operational defect management and constitutes an enterprise governance risk.

Rather than evaluating recalls as isolated technical failures, COSO enables assessment of systemic exposure across five interrelated components: Governance & Culture, Strategy & Objective-Setting, Performance, Review & Revision, and Information & Communication. This structure ensures that statistical findings are interpreted in terms of enterprise risk oversight and board-level accountability.

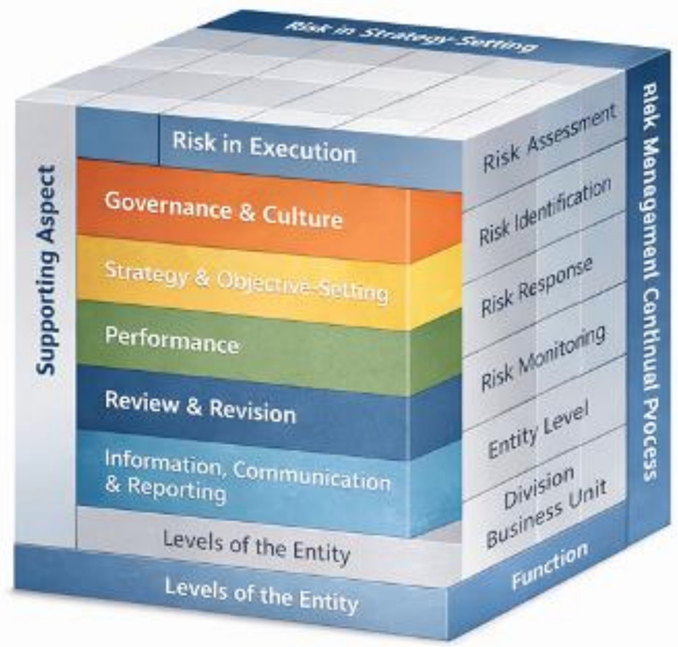

Figure 1 COSO Enterprise Risk Management (ERM) Framework (2017).

Source 1 Committee of Sponsoring Organizations of the Treadway Commission (COSO), 2017.

COSO-to-Analytics Mapping

To operationalise the COSO framework within a quantitative risk analysis context, recall exposure was transformed into a severity-weighted Risk Index. The Risk Index is calculated by multiplying the number of vehicles affected by the assigned severity weight.

These metric shifts emphasise from recall frequency to material exposure. By integrating advisory severity with affected vehicle volume, the Risk Index aligns recall analysis with enterprise-level impact rather than campaign counts.

Each COSO component is supported by specific analytic outputs. The mapping below summarises how each COSO component is operationalised through specific analytical outputs and quantitative evidence.

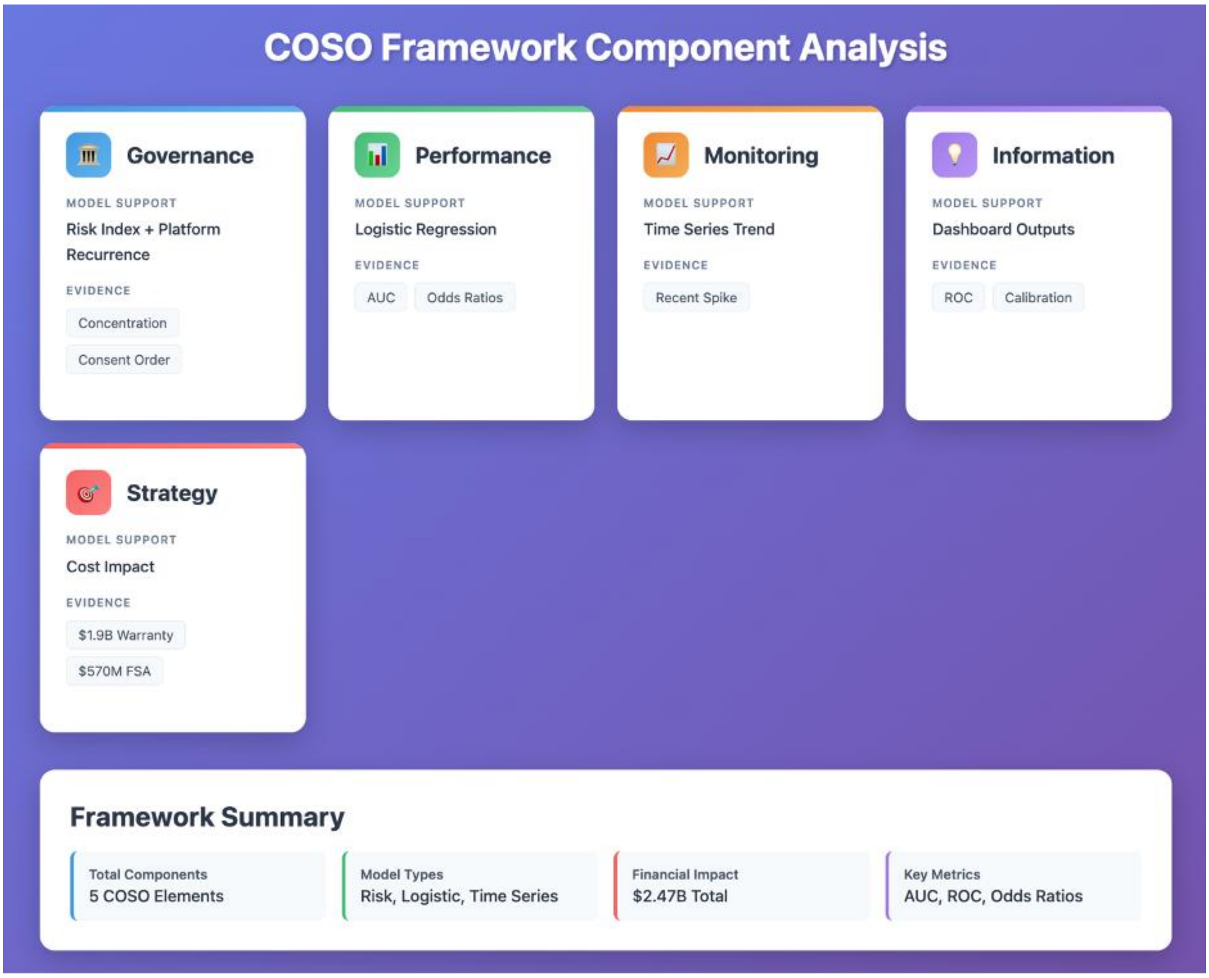

Figure 2 COSO Framework Component Analysis

This mapping ensures that statistical diagnostics are interpreted within an enterprise governance structure rather than as isolated modelling outputs.

Alternative Frameworks Considered

Several alternative governance and performance frameworks were evaluated before selecting COSO ERM as the primary analytical lens. The ISO 31000 (International Organization for Standardization, 2018) offers more recognised risk guidance, although it is more procedural and less clear in its correlation of risk with strategy and board oversight. The Balanced Scorecard (Kaplan and Norton, 1996) is concerned with performance measurement as opposed to the balanced risk in governance and operationalization tools like DMAIC and COPQ are concerned with the process and cost problems but not at the enterprise level. Since Ford carries a regulation and governance risk, COSO ERM was chosen as the main prism since it incorporates governance, strategy, performance, monitoring, and reporting under one umbrella (Committee of Sponsoring Organizations of the Treadway Commission [COSO], 2017).

Analytical Approach

We explored the structured analytics frameworks such as CRISP-DM and the Kimball Methodology, but they emphasise full deployment and organisational process integration beyond the scope of this consultancy analysis (hearer, 2000, pp. 15–18; Kimball & Ross, 2013, pp. 6–12). Because the objective was rapid risk identification and prioritisation from existing recall data rather than building a production system, a focused data-mining approach was adopted to support efficient exploration and predictive modelling.

Analytical Methods

Data mining contains more than one method that can be used to solve Ford`s quality defects. According to Shmueli et al. (2016), "Data mining refers to business analytics methods that go beyond counts, descriptive techniques, reporting, and methods based on business rules." (p. 245). We have used more than one model for our analysis. However, to understand the problem more deeply, we explored descriptive analysis.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) served as the structural foundation of the analytical framework. Its objective extended beyond descriptive reporting to the discovery of exposure architecture embedded within recall behaviour. In automotive recall systems, raw campaign frequency is an insufficient measure of enterprise risk because a single event can affect millions of vehicles. The analytical strategy, therefore, redefined recalls as exposure-driven phenomena rather than isolated incidents.

The cleaned dataset was aggregated at the campaign level and enriched with component classifications and severity indicators. A severity-weighted exposure metric was constructed to convert heterogeneous recall records into a unified quantitative risk surface. The risk index was defined as:

$$Risk\ Index = \sum (Affected\ Vehicle\ \times\ Severity\ Weight)$$

where severity weights encode regulatory escalation. Campaigns labeled Do Not Drive were assigned a weight of 3 to represent maximum hazard exposure, Park Outside advisories received a weight of 2 to reflect elevated but non-critical risk, and standard recalls retained a baseline weight of 1. This ordinal calibration transforms qualitative advisory language into a consistent numerical scale, enabling cross-campaign comparison without assuming equal impact across recall types. The objective of this transformation was to shift analytical interpretation from event frequency toward the magnitude of exposure.

Temporal aggregation was performed at yearly levels to examine structural drift in recall exposure. Preliminary trend analysis demonstrated that recall behaviour is episodic rather than linear. A small number of extreme campaigns dominate cumulative impact, producing a heavy-tailed distribution in which upper-tail events govern enterprise vulnerability. Recognising this asymmetry is analytically important because average-based interpretation obscures risk concentration and underestimates catastrophic exposure.

Component-level aggregation was then used to evaluate concentration across Ford's technical architecture. Severity-weighted exposure was aggregated at the subsystem level and ranked by cumulative contribution to total enterprise risk using a Pareto-style distribution analysis. Heatmap visualisation extended this aggregation by mapping component intensity over time and across vehicle platforms. Rather than treating recalls as isolated mechanical failures, this approach reveals persistent vulnerability clusters embedded within shared system architectures.

A priority matrix integrated recall frequency and exposure magnitude into a geometric representation of enterprise risk. Each component was positioned in a two-dimensional space where the horizontal axis captures recurrence, and the vertical axis captures severity-weighted impact. Bubble size encoded total affected vehicles to preserve magnitude information. Quadrant boundaries were defined using high-percentile thresholds (85%) instead of median splits to isolate extreme exposure zones consistent with heavy-tailed behaviour. The resulting structure distinguishes chronic operational friction from catastrophic outliers and reframes recall behaviour as a spatial risk surface.

Platform linkage analysis extended the structural perspective by mapping component exposure onto shared vehicle architectures. Risk amplification occurs when vulnerable systems intersect with high-volume production platforms, converting moderate engineering defects into enterprise-scale consequences. Enterprise exposure, therefore, emerges from the interaction between engineering reliability and production concentration rather than from isolated component failures.

Throughout the analytical process, visualisation was treated as an instrument of structural discovery. Trend curves, Pareto concentration, heatmaps, and priority matrices were selected to expose distinct properties of the recall system: temporal volatility, concentration asymmetry, architectural persistence, and platform amplification. Together, these methods define the operational geometry of enterprise risk prior to predictive modelling.

Model 1 (Binary Logistic Regression)

Binary logistic regression is a probabilistic classification method used to estimate the likelihood that an observation falls into one of two outcome categories (Dick et al, 2022). In this report, it was applied to classify recall events as High Impact or Not High Impact by generating a probability score that reflects estimated risk. This score is then compared to a threshold to produce a final category, enabling both risk ranking and clear decision support. The model examines how multiple factors, including severity indicators and recurrence patterns, jointly influence the probability of a high-impact outcome, while remaining transparent and interpretable for executive audiences.

This model was selected because the objective was to prioritise recall cases by business impact rather than diagnose technical root causes. Logistic regression converts complex variables into a single comparable probability measure. It allows management to quickly identify which recalls warrant immediate attention. The High-Impact label was defined using a severity-weighted warranty index threshold (above 75%). It enables the model to function as a structured risk-triage tool that supports early warning and resource allocation decisions.

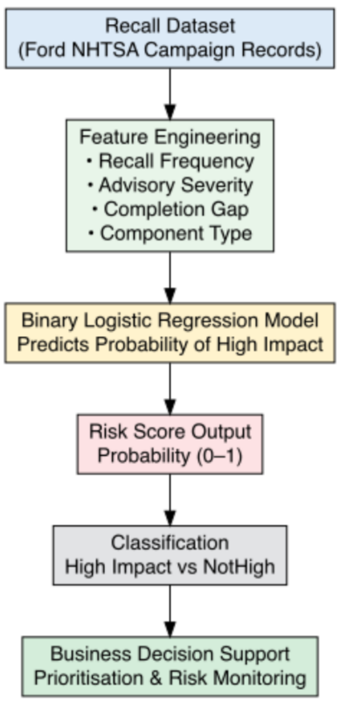

Figure 3 Binary Logistic Regression Model Framework

The analysis was conducted in R, with tidyverse, caret, pROC, and ggplot2 to support data transformation, model training, validation, and visualisation. Before modelling, recall records were cleaned and integrated across the NHTSA recall file and the campaign-level file, with standardised formats and a join completeness check. To ensure stable numeric inputs, key fields such as potentially affected units and completion rate were parsed using a robust numeric conversion function that handles common reporting formats (e.g., commas, percentages, "Not Reported"), while advisory indicators were converted into consistent binary flags. Missing values in potentially affected units and completion rate were imputed using dataset medians, and a completion gap was derived to capture the remaining exposure from incomplete remedies. Severity was operationalized through an advisory severity score based on "do not drive" and "park outside" indicators, and a recall-type multiplier was applied to reflect differences between vehicle, equipment, tyres, and child-seat recalls. To capture recurrence risk, we engineered count-based features for how often the same platform, component, and platform–component pairing appears in the dataset, then log-transformed those counts for modelling stability. Finally, high-cardinality identifiers were bucketed into the most frequent categories (with the remainder grouped as "OTHER") to keep the model interpretable and robust. Following data preparation, 6009 recall records remained for analysis. The model was then trained on an 80/20 train–test split (4008/1201 datapoints) using a fixed random seed for reproducibility. Predictors included log recurrence measures (platform, component, and their interaction), advisory severity, and the bucketed platform/model-year and component groupings. The fitted model outputs a predicted probability of high impact for each case, which was then converted into a classification using a data-driven threshold selected from ROC analysis (Youden's method).

Assumptions and limitations

The logistic regression model operates under several analytical assumptions that shape how its outputs should be interpreted. Recall events are treated as independent observations, although shared manufacturing cycles or supply chains may introduce hidden relationships that influence probability stability. The model also assumes consistency in reporting standards and severity classifications over time, as changes in documentation practices can shift observed risk distributions and affect calibration behaviour. Also, the engineered severity indicators and selected probability threshold are assumed to reasonably represent what constitutes a high-impact recall. Because the model produces probabilities rather than certainties, its value lies in directional risk ranking rather than definitive prediction, reflecting a managerial preference for conservative escalation over exhaustive detection.

The model also has some limitations. While it demonstrates moderate discrimination (AUC = 0.644) and reasonable classification accuracy (0.669), its predictive strength is not absolute. Sensitivity of 0.487 indicates that some high-impact recalls may remain undetected, whereas higher specificity (0.729) shows stronger performance in identifying lower-risk cases. Calibration analysis further indicates that probability reliability varies across bands (below 20% and above 25%) due to sparse event density. Logistic regression is also limited in capturing complex non-linear interactions, meaning the analysis identifies risk patterns rather than engineering root causes. For these reasons, the model is most effective as an early-warning and prioritization tool within a broader analytical framework rather than a standalone decision authority.

Model 2 (Classification Tree – Random Forest)

One of the data mining techniques is a predictive model, for the analysis to identify the possible recall reason, the classification tree is the method to be used (Shmueli et al., 2016). In this method, observations are separated into subgroups by creating splits on predictors (Shmueli et al., 2016).

Shmueli et al. (2016) explain the Random Forest algorithm:

> The basic idea in random forests is to draw multiple random samples, with replacement, from the data (this sampling approach is called the bootstrap). Fit a classification (or regression) tree to each sample (and thus obtain a "forest"). Combine the predictions/classifications from the individual trees to obtain improved predictions. Use voting for classification and averaging for prediction (p. 202).

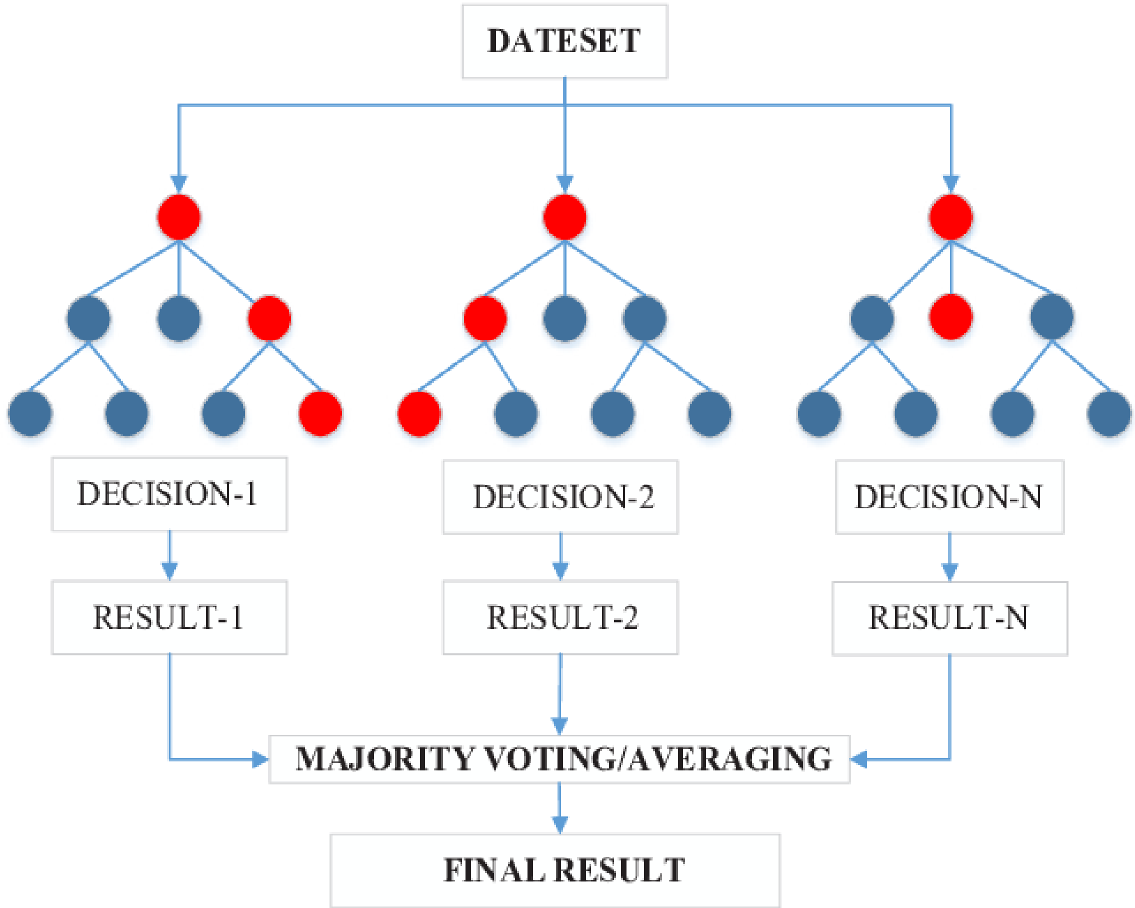

Figure 4 Random Forest Model Tree Image

With these concepts clear, we can proceed to the actual analysis using the Random Forest model. This model is best suited for the classification of the new recalls, because of the following reasons: Non-linear: Handles complex text patterns; High-Dimensional: Works with 202 features; Imbalance-Robust: Manages 137 classes; Ensemble Power: 300 trees with one 1 model (tree limit depth of 15) and prepared to deal with imbalanced data; Noise-Robust: Handles messy real-world text. The method Multinomial Logistic Regression Model was also considered to be used, but the results (In the Finding section) show that the best model for this sample is the Random Forest Model.

The coding language used for this model is Python, with the following libraries: scikit-learn 1.0.2. for machine learning, pandas 1.4.2, numpy 1.22.3, matplotlib 3.5.1 and seaborn 0.11.2 with text preprocessing (Term Frequency-Inverse Document Frequency).

Lastly, the data preparation was done based on the 7498 recall data points. 843 data points were deleted due to missing or empty data in the summary, consequence or remedy. 80 data points were deleted due to representing ultra-rare cases (Less than 3 cases in the total sample). The last filter used was the time. The chosen samples were from the year 2015 until 2026, which reduced the sample size to 2005 data points.

In these 2005 samples, text fields were concatenated (summary, consequence and remedy), then a pareto of 80/20 (1604/401 of the 2005 recall data points) were used to segregate between training and test samples.

Assumptions and limitations

One of the limitations of this analysis is that the data collected is from a secondary source, which means that the actual data from Ford's company can also be used to enhance even more the models that are discussed in the report.

For the model, some of the limitations Random Forest model are: Less interpretable: harder to explain exact drivers compared to Logistic Regression, risk of overfitting: Performance may reduce on unseen future recall types, computationally heavier: Requires more resources and tuning for deployment, does not capture external changes: Supplier shifts, regulatory updates, or new technology risks are not included.

Findings

In this report, multiple analyses were deployed in a way to address multiple fronts of Ford's Quality issues. It was identified which components are the critical ones in the short term, including a probabilistic risk-classification model (Model 1 – Binary Logistic Regression) used to estimate which recall events are most likely to escalate into high-impact cases and therefore require immediate managerial attention, and a model to classify the recall component failure based on the text description (Model 2 – Random Forest). However, before modelling, the analysis first examined the recall landscape through exploratory data analysis (EDA) and a structured risk-management framework to establish context and identify emerging patterns.

Framework Findings (COSO-aligned Results)

Governance & Culture – Structural Risk Concentration

The severity-weighted Risk Index demonstrates that recall exposure is not evenly distributed across Ford's portfolio. Approximately 70–80% of total exposure is concentrated within a limited number of platforms–component clusters. This concentration indicates structural risk rather than random dispersion.

Logistic regression analysis reinforces this pattern. The F-350 SD (2022) platform exhibits approximately 5.7× higher odds of high-impact escalation (statistically significant), while platform recurrence behaviour increases escalation probability by 1.3×. These findings suggest recall vulnerability is embedded within specific platform ecosystems rather than driven by isolated defects.

Under COSO, such clustering elevates recall exposure to a governance-level concern requiring platform-level oversight.

Strategy & Objective-Setting – Material Enterprise Impact

Recall exposure directly affects Ford's strategic objectives through warranty volatility (>$1B pressure), regulatory scrutiny (NHTSA consent order), and brand risk.

The logistic model (AUC = 0.644) demonstrates meaningful discrimination between high- and lower-impact recalls, indicating identifiable structural drivers rather than random variation. Although high-impact recalls represent a minority of total campaigns, they account for a disproportionately large share of severity-weighted exposure.

This reframes recall management as a strategic prioritisation issue rather than a compliance-only function.

Performance – Escalation Drivers

Marginal-effects analysis identifies platform–component interaction as the strongest driver of high-impact probability (+6.1 percentage points). Platform recurrence alone remains statistically significant, whereas component recurrence independently shows a limited effect.

Model sensitivity (0.487) and specificity (0.729) indicate conservative classification behaviour suitable for governance triage rather than automated decision-making.

Collectively, these results confirm that recall exposure is structurally concentrated within specific vehicle platforms rather than randomly dispersed.

Review & Monitoring – Exposure Persistence

Time-series diagnostics indicate that severity-weighted exposure remains concentrated in recent periods rather than stabilising. Calibration analysis shows slight underestimation in higher-risk bands, suggesting escalation potential may be understated in certain clusters.

Under COSO's Review & Revision component, this supports embedding Risk Index tracking and probability scoring into governance dashboards to enable earlier detection and intervention.

Information & Communication – Governance Tools

Recall exposure is operationalised through: Severity-weighted Risk Index, Escalation probability scoring (Logistic Regression), Platform-level risk multipliers, and Random Forest classification (95.5% accuracy)

The Random Forest model reduces estimated investigation time from 20 days to 1.5 days (≈92.5% reduction under stated assumptions), strengthening governance responsiveness and resource allocation efficiency.

These tools convert retrospective reporting into structured enterprise risk intelligence.

Alternative Interpretation

An alternative explanation is that exposure concentration reflects product mix scale rather than structural governance weakness. However, statistically significant platform recurrence effects (1.3×) and 5.7× platform multipliers suggest systemic exposure embedded within specific platform ecosystems rather than scale effects alone.

Across all analytical layers, Ford's recall exposure is concentrated, platform-driven, and strategically material. Under COSO ERM, the evidence indicates with statistical support that recall exposure operates as an enterprise governance risk requiring structured platform-level oversight, predictive monitoring, and integrated risk reporting.

Exploratory Data Analysis (EDA)

In EDA, unless otherwise stated, component-level structural analyses focus on the 2021–2025 period in order to capture recent exposure patterns, while long-horizon trend analysis uses the full 2010–2025 dataset. Severity-weighted recall exposure reveals a structural escalation in enterprise risk over the observation period. While recall frequency increases gradually, the magnitude of affected vehicles expands disproportionately. Enterprise vulnerability is driven more by scale than by count. Recalls have evolved from routine operational disturbances into episodic high-impact shocks capable of affecting millions of vehicles simultaneously. The exposure system exhibits discontinuities rather than smooth growth.

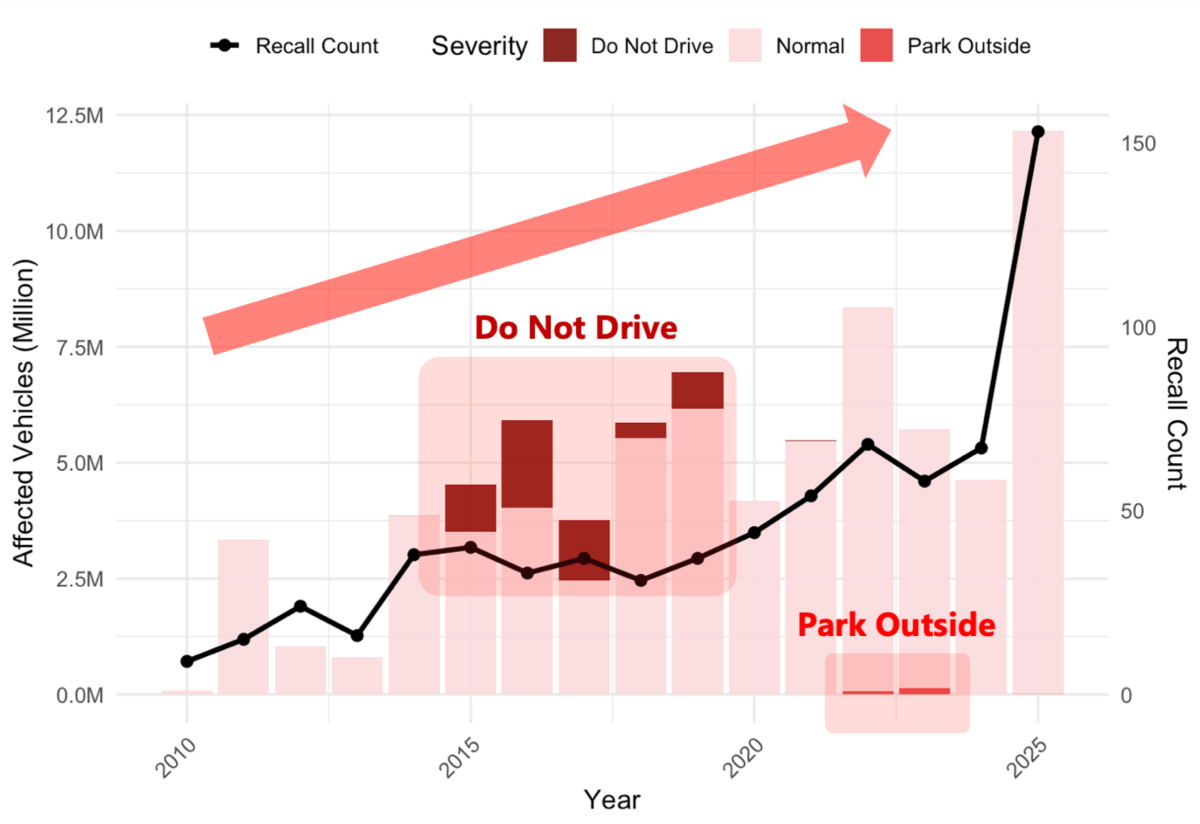

Figure 5 Recall Count and Affected Vehicle

The temporal exposure curve demonstrates that enterprise risk is governed by episodic spikes associated with regulatory escalation campaigns. Advisory categories such as Do Not Drive and Park Outside disproportionately influence cumulative exposure because their severity weighting amplifies impact. The distribution is heavy-tailed: rare upper-extreme events dominate long-run enterprise risk. Average recall behavior is, therefore, a poor proxy for vulnerability. Structural exposure emerges from tail concentration rather than central tendency.

Forecast modelling reinforces this instability. Both ETS and ARIMA projections indicate that recall exposure is unlikely to return to historical baselines. Instead, the forecast envelope widens over time, signaling increasing volatility rather than convergence. The system is becoming more shock-sensitive as software-electronic integration and platform complexity expand. Forecast uncertainty is therefore not a modelling artifact but an empirical property of the recall system.

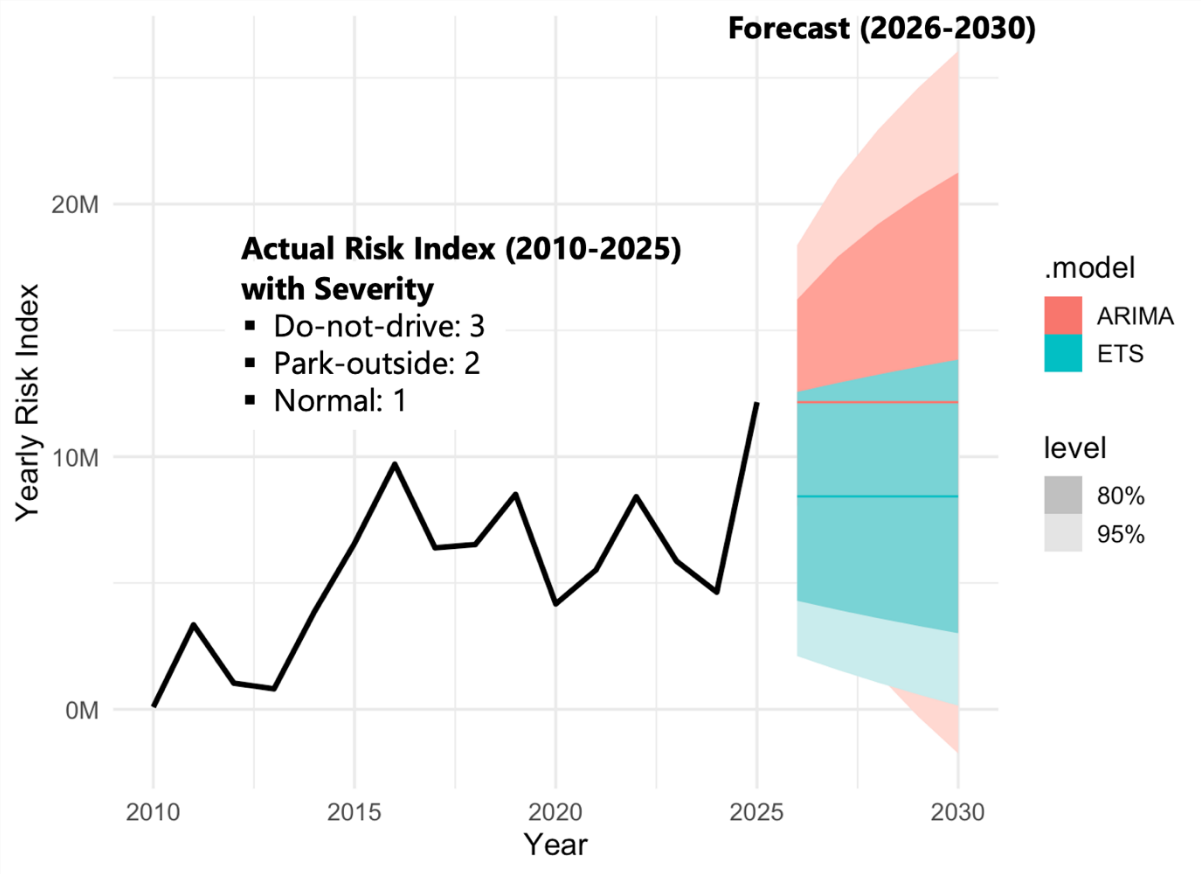

Figure 6 Yearly Risk Index Forecast

Within the 2021–2025 exposure window, component aggregation reveals pronounced concentration. A narrow cluster of subsystems accounts for a disproportionate share of enterprise exposure. As illustrated in the Pareto structure, sensing systems and software dominate the upper tail, followed by airbags and automatic transmissions. These components behave as architectural bottlenecks rather than isolated anomalies. Enterprise vulnerability is concentrated rather than evenly distributed.

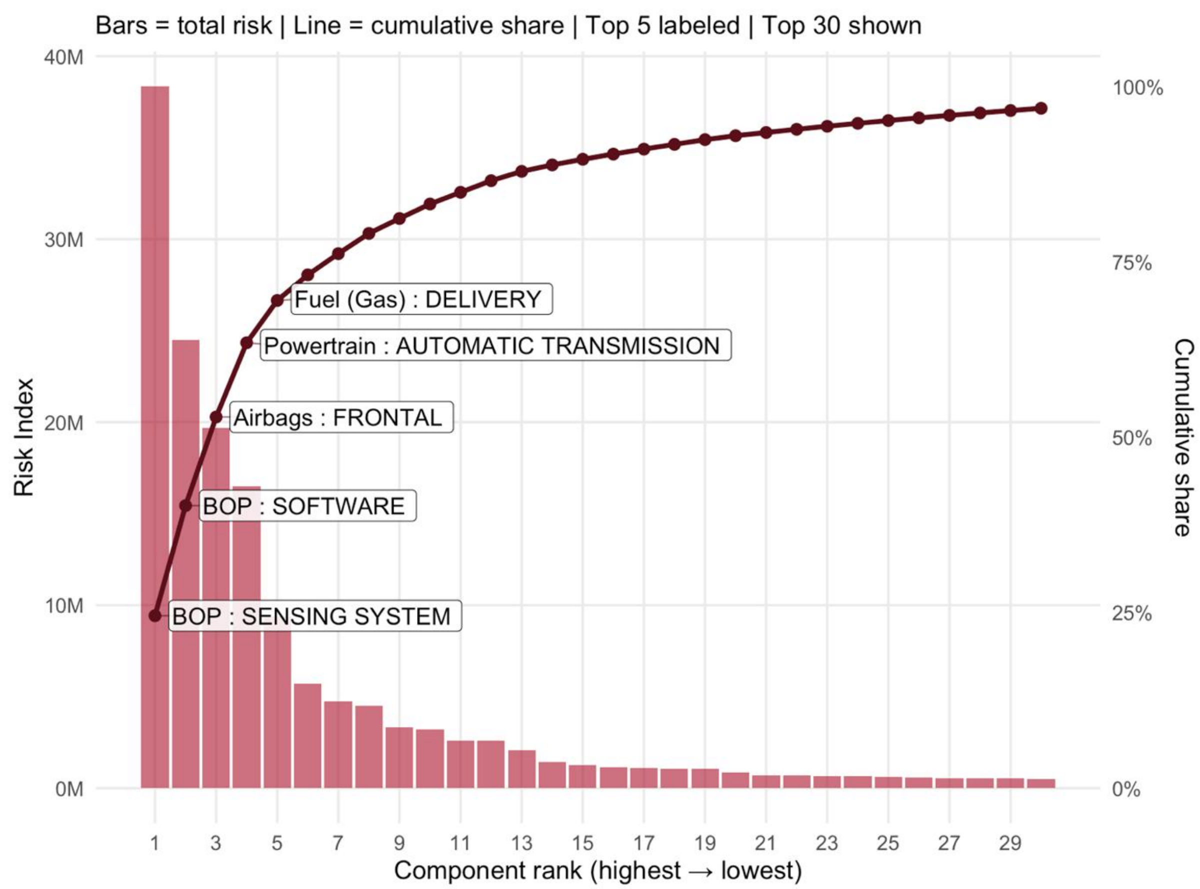

Figure 7 Pareto of Severity-weighted Risk (2021-2025)

The 2021–2025 heatmap clarifies both persistence and directional shift. High-risk components recur across multiple years, indicating embedded structural fragility. However, recent exposure is increasingly dominated by software and sensing integration. Earlier recall patterns were mechanically distributed; current exposure is digitally concentrated. Software failures propagate horizontally across shared architectures, affecting multiple platforms simultaneously. This marks a transition from localized hardware failures to systemic digital vulnerability.

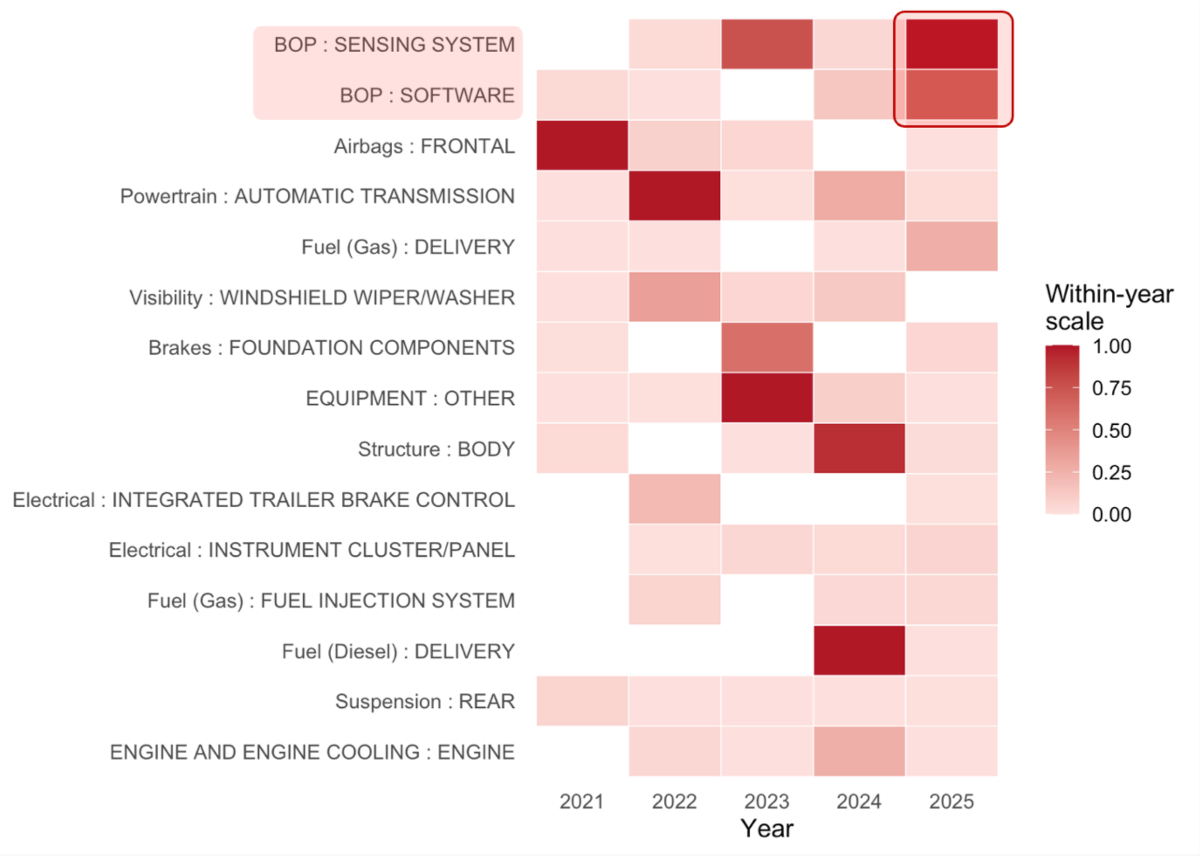

Figure 8 Heat Map Component Risk (2021-2025)

The priority matrix constructed from 2021–2025 aggregated exposure. Components in the upper-right quadrant combine high recurrence with high severity and therefore function as compounding risk multipliers. Sensing systems, software, airbags, and automatic transmissions cluster within this critical zone. These components propagate enterprise risk rather than merely contributing to it. In contrast, low-frequency or low-impact components occupy regions that represent operational noise rather than structural threat.

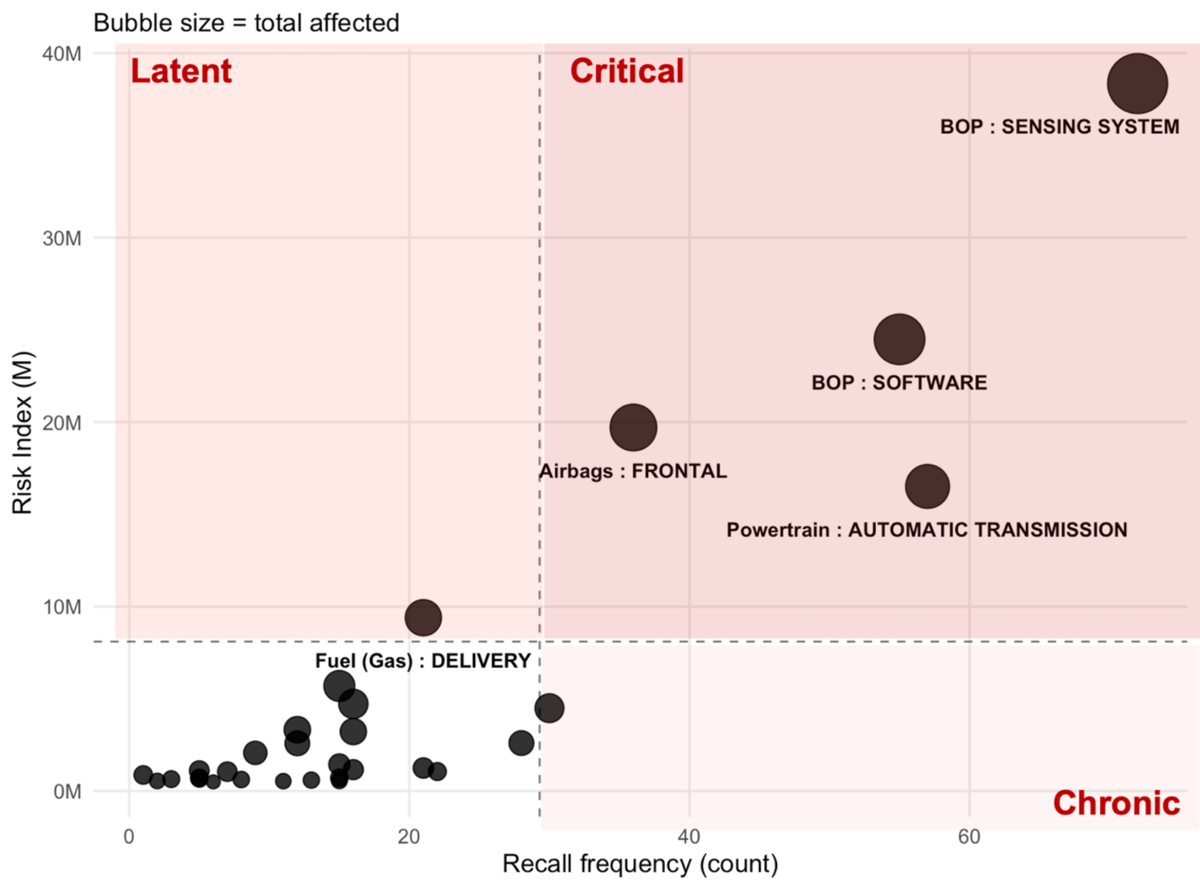

Figure 9 Component Priority Matrix (2021-2025)

Platform mapping (2021-2025) exposes amplification mechanisms embedded in shared architectures. The component × model heatmap shows that sensing and software subsystems span nearly all major vehicle lines. Their cross-platform presence enables a single defect to diffuse exposure across a large installed base. Airbags and transmissions, by contrast, display platform-specific clustering, indicating localized engineering stress points. Enterprise exposure therefore emerges from layered interaction: shared systems distribute risk horizontally, while concentrated platforms intensify risk vertically.

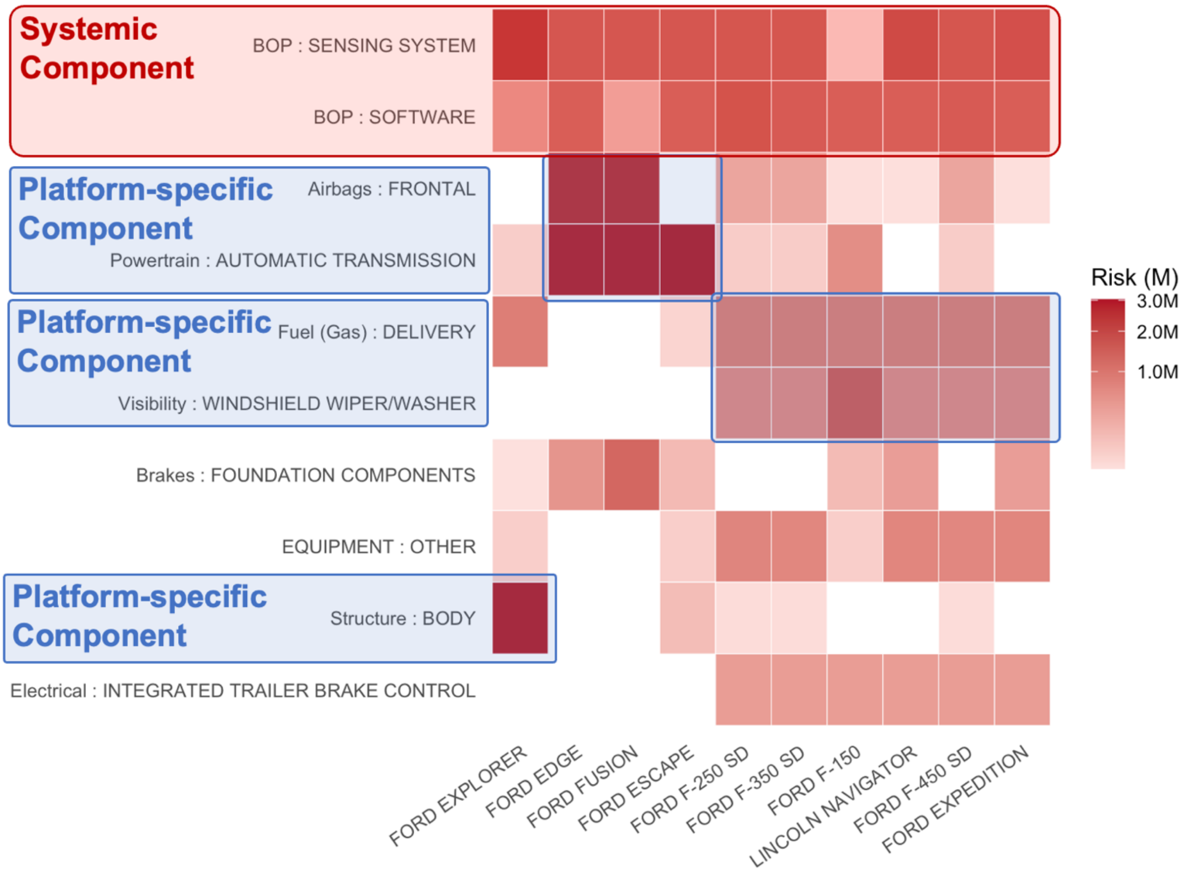

Figure 10 Component x Model Heatmap (2021-2025)

Taken together, the findings demonstrate that recall exposure is concentrated, heavy-tailed, persistent, and increasingly software-driven. Enterprise vulnerability arises from architectural coupling between digital systems and high-volume platforms. The figures collectively reveal that recall behaviour is structural rather than random, governed by recurring exposure geometries that define the system's risk architecture.

Model 1 (Binary Logistic Regression)

Model Performance

For Model 1, model discrimination was assessed using a Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC), which together evaluate how effectively the model separates high-impact recalls from non-high-impact recalls across all possible decision thresholds (Hoo et al., 2017).

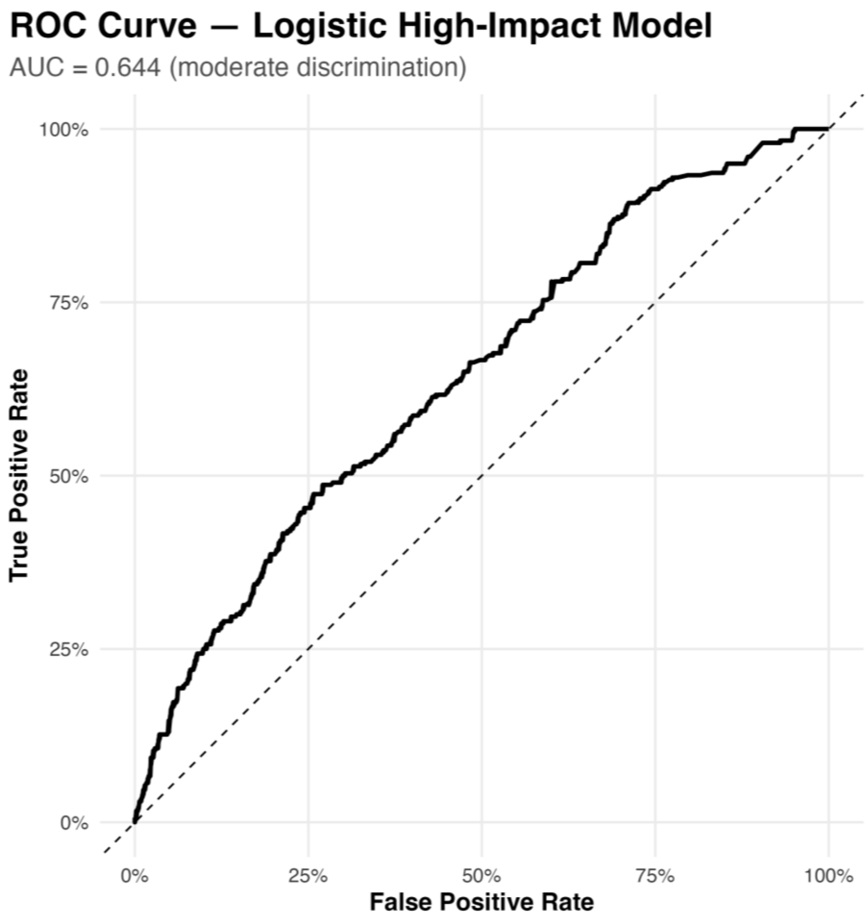

Figure 11 ROC/AUC curve for Model 1

On the test dataset, the model achieved an AUC of 0.644, indicating moderate discrimination ability. In practical terms, this means the model is meaningfully better than random guessing at ranking recalls by risk severity, making it suitable for prioritization and triage, though not for definitive classification without human oversight.

To understand classification performance at a specific operating point, results were then evaluated using a confusion matrix. It is a diagnostic table that compares predicted versus actual outcomes to show how many cases were classified correctly or incorrectly (Chicco et al., 2021).

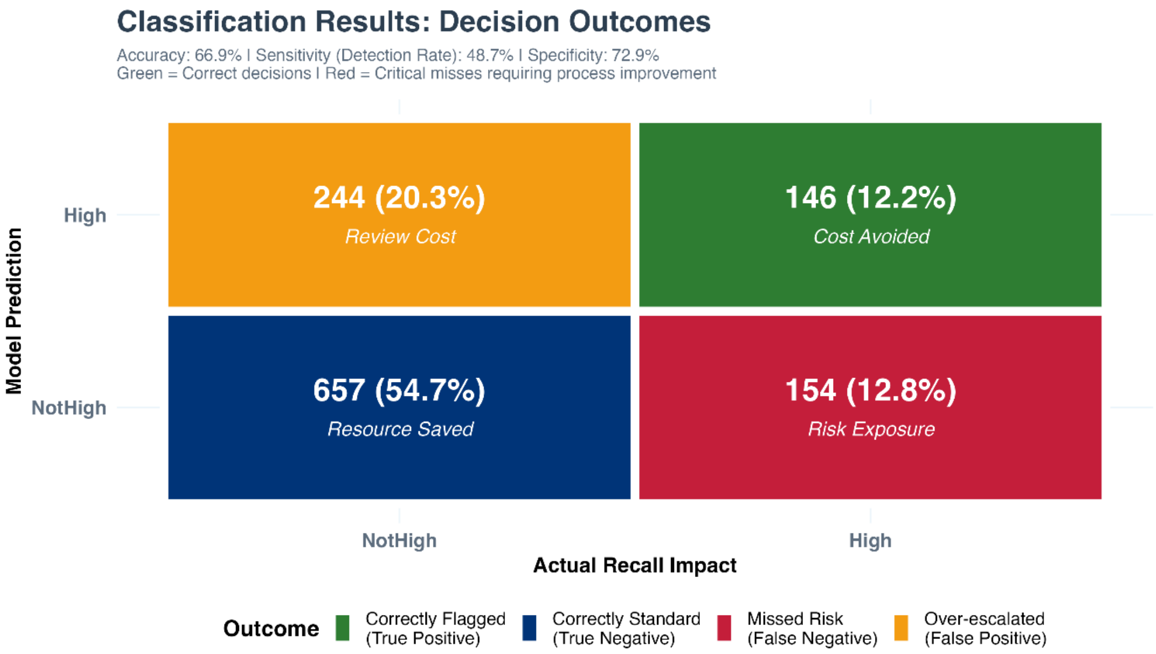

Figure 12 Confusion Matrix for Model 1

In this analysis, it illustrated how effectively the model distinguished High-Impact from Not-High recalls and highlighted the trade-off between missed severe cases and false alarms. At the selected probability threshold, overall accuracy was 0.669, meaning approximately two-thirds of recalls were correctly classified. The model achieved a sensitivity (recall for high-impact cases) of 0.487, indicating that just under half of true high-impact recalls were successfully detected, and a specificity of 0.729, showing stronger performance in correctly identifying lower-risk recalls. The model is more conservative in labelling recalls as high risk, which reduces false alarms but increases the chance of missing some severe cases. Such a balance is appropriate when predictions are intended to support managerial judgement rather than replace it.

Beyond classification accuracy, the reliability of predicted probabilities themselves was assessed through a calibration analysis, which compares predicted risk percentages with actual observed high-impact rates (Fieberg et al., 2018).

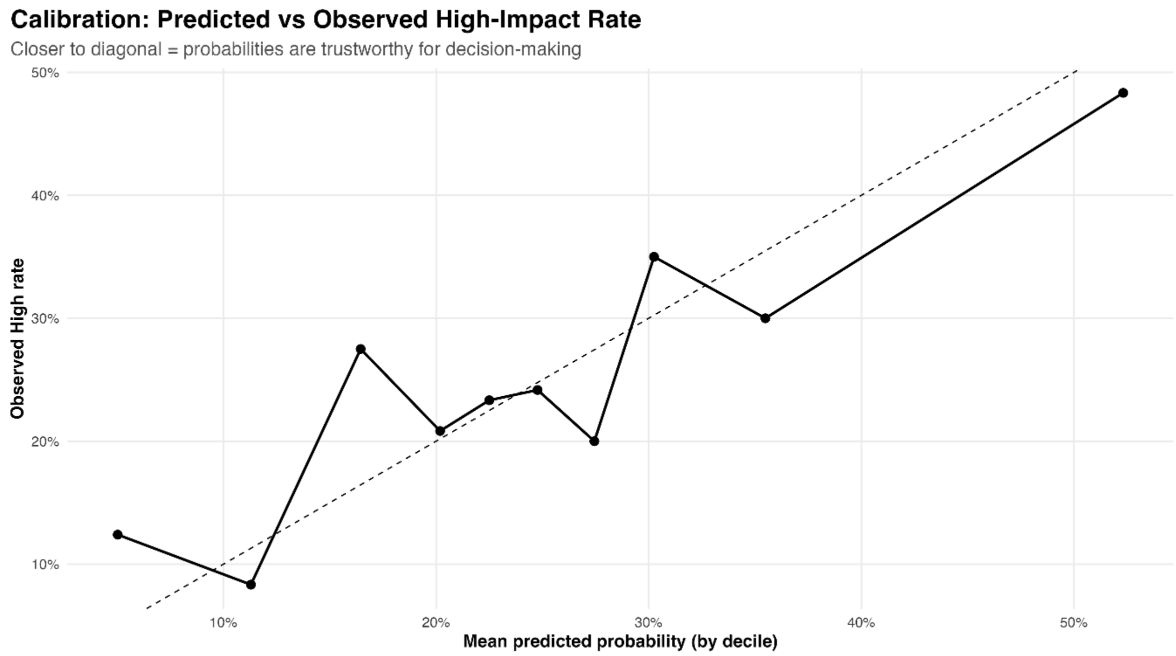

Figure 13 Calibration plot for Model 1

The calibration curve demonstrated very close alignment in the 20–25% probability band, indicating strong reliability for mid-level risk estimates. At higher probability levels above approximately 35%, the model showed a tendency to underestimate true risk, meaning actual high-impact occurrence was slightly higher than predicted. In the lowest probability deciles (around 5–12%), noticeable fluctuation was observed, reflecting greater uncertainty when predicted risk is very low. Overall, calibration was directionally acceptable, suggesting that probability outputs are trustworthy as relative risk indicators, although they should be interpreted as guidance rather than precise forecasts.

Model Interpretation and Business Insights

While the previous section evaluated whether the model performs reliably from a statistical perspective, the following visuals explain what the model is revealing about recall risk behavior and managerial prioritization.

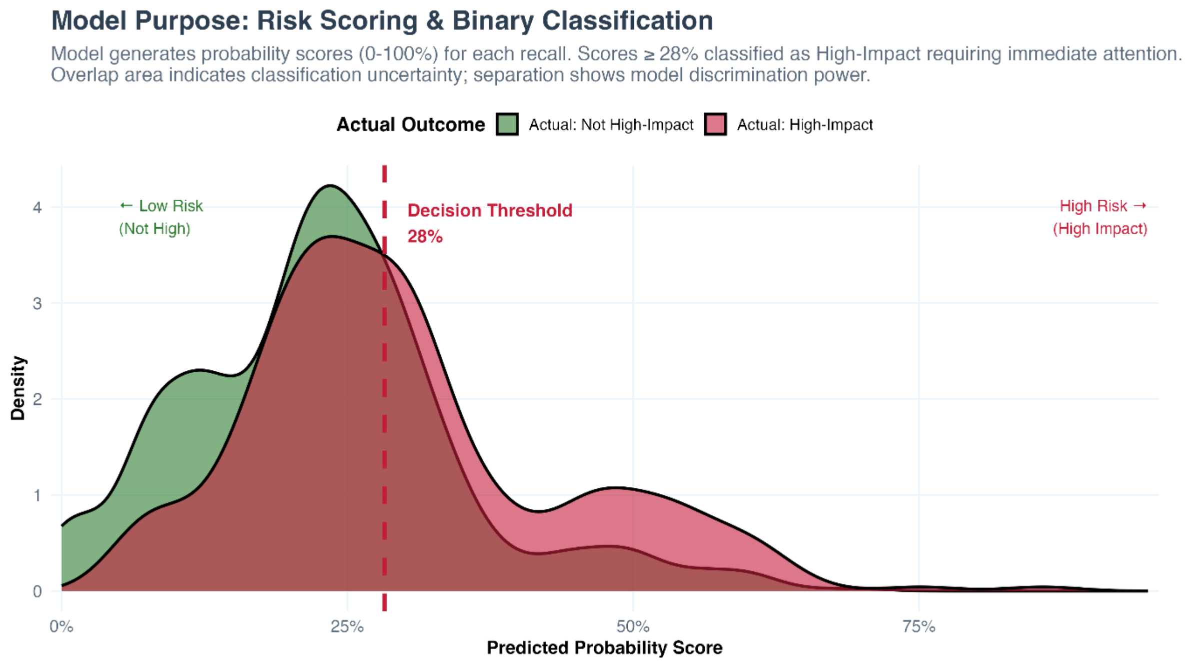

Figure 14 Risk Scoring and Binary Classification

This visual presents the distribution of predicted probability scores for recalls that were ultimately classified as High-Impact versus Not-High. The horizontal axis represents the model's estimated likelihood of escalation, while the vertical density curves illustrate how frequently each probability range occurs for the two actual outcome groups. The dashed vertical line at 28 percent marks the operational decision threshold used to classify recalls as requiring immediate attention.

The overlapping region between the two curves highlights the natural uncertainty inherent in real-world classification tasks. In this zone, both high-impact and lower-impact recalls share similar probability scores, which explains the presence of false positives and false negatives observed in the confusion matrix. Conversely, at very low probability levels the Not-High curve dominates, and at higher probability ranges the High-Impact curve becomes more prominent, indicating clearer separation. From a managerial standpoint, this visual demonstrates that the model functions primarily as a risk-scoring and triage mechanism rather than a binary verdict, enabling prioritized review rather than automated escalation.

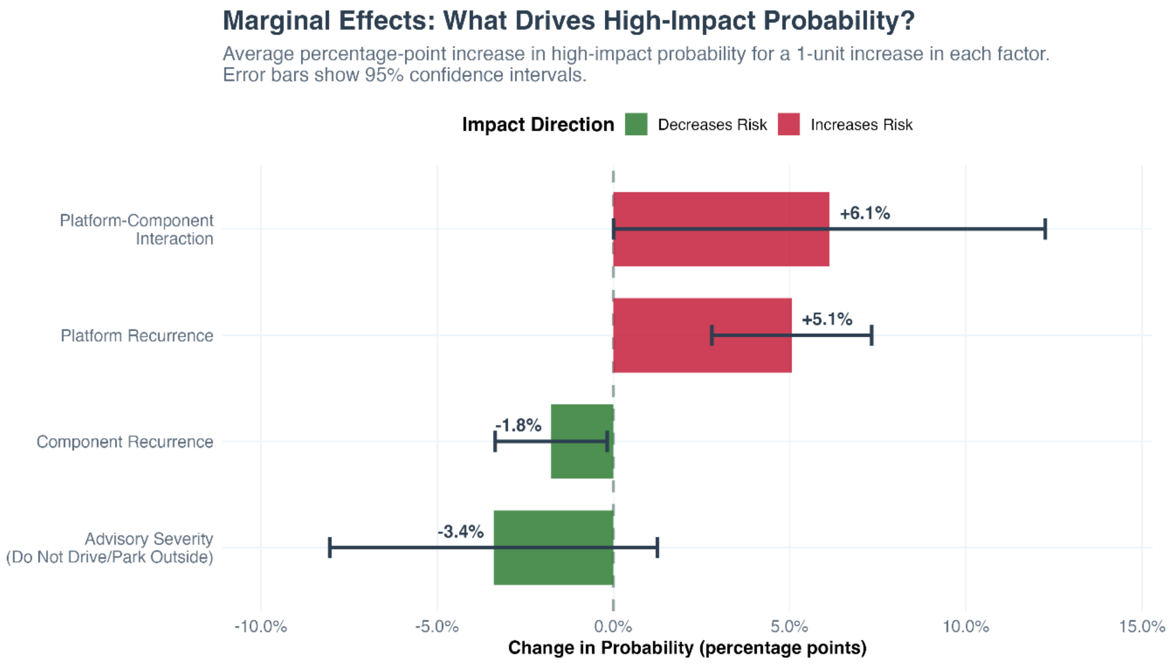

Figure 15 Drivers of High-Impact Probability

The marginal-effects chart quantifies how incremental changes in each analytical factor influence the predicted probability of a high-impact recall, holding other variables constant. Bars extending to the right indicate increased risk probability, while bars to the left represent reductions. Error bars denote 95 percent confidence intervals, which show statistical stability or uncertainty.

The marginal-effects results indicate that the platform–component interaction produces the largest increase in high-impact probability (+6.1%). It means risk rises most when the same component failures recur within a specific vehicle platform. The platform factor alone is the second strongest (+5.1%) and shows narrow confidence intervals. This confirms it as a stable and significant driver of escalation. In contrast, component recurrence on its own slightly decreases probability, while advisory severity indicators show minimal influence. For management, this means the vehicle platform itself is the main factor in driving risk, especially when the same components keep failing. Monitoring platform-level patterns will therefore have the biggest impact on preventing serious recalls.

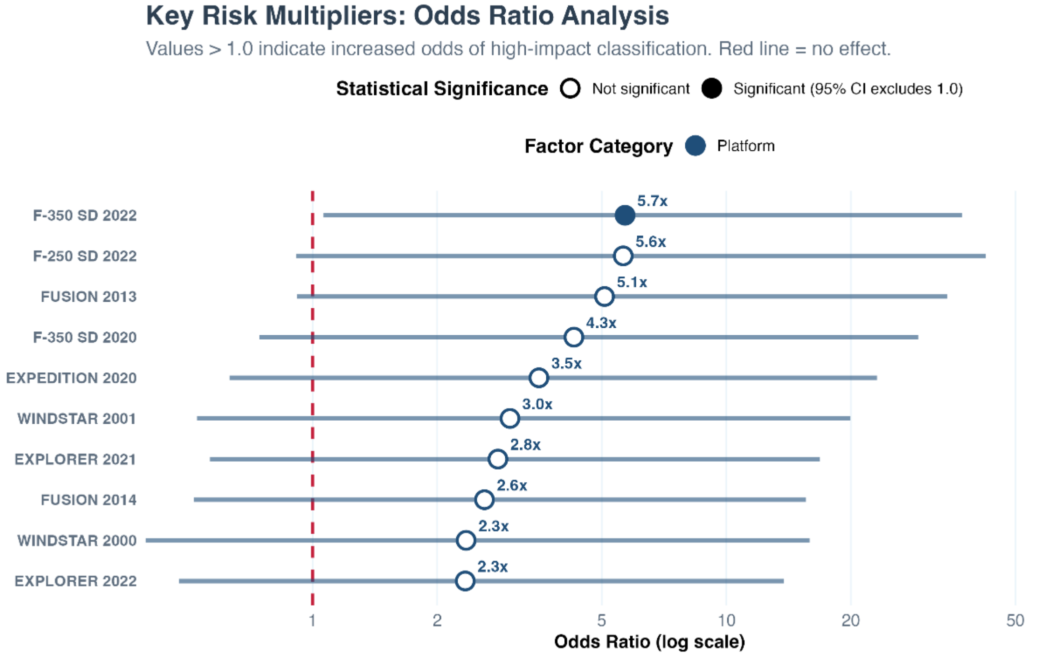

Figure 16 Key Risk Multipliers (Top 10)

This chart shows the top 10 risk multipliers, all of which are platform related. It also aligns with Figure 15 where platform behavior also emerged as the most influential driver of escalation probability. It expresses risk multipliers using odds ratios for specific platform and model-year categories. Values positioned to the right of the neutral reference line (odds ratio = 1) indicate increased odds of being classified as high impact. The horizontal lines represent confidence intervals, with filled markers indicating statistical significance.

The F-350 SD (2022) platform shows an approximately 5.7× higher likelihood of high-impact escalation, and this result is statistically significant. It means the effect is consistent and unlikely to be due to random variation in the data. However, the width of many confidence intervals indicates varying degrees of uncertainty. It means these multipliers should be interpreted as priority signals for investigation rather than precise forecasts. From a governance perspective, the chart enables management to identify which platform cohorts warrant deeper engineering or supplier audits, while recognizing that statistical variability prevents deterministic conclusions.

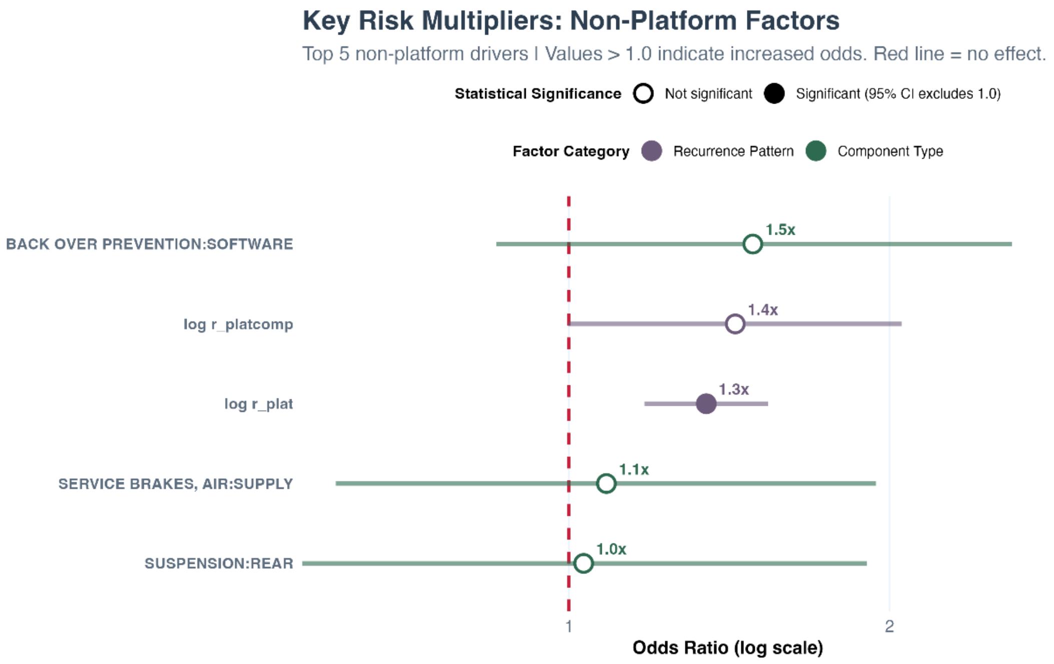

Figure 17 Key Risk Multipliers (Non-Platform Factors)

Although this chart excludes specific platform names, it still shows that platform recurrence behavior remains influential because the model captures how often the same platform appears in recalls, not only its identity. In practical terms, repeated platform patterns alone increase the likelihood of escalation by around 1.3× (statistically significant). It indicates that risk is embedded in structural design cycles rather than isolated incidents.

Component variables then become the next most useful layer of insight, with several categories raising high-impact odds by roughly 1.0–1.5× (though not statistically significant). This means that once a risky platform pattern is recognized, component type helps pinpoint where technical issues are most likely to arise. For management, it is important to understand that platform drives the probability while the components refine the diagnosis for managerial action.

Key Executive Insight

The model indicates that recurring platform patterns are the strongest signal of future high-impact recalls, with platform–component interactions increasing escalation likelihood by up to 5–6× compared to baseline cases. For strategic action, this means quality interventions should prioritize platform-level design and supplier ecosystems first, then use component-level patterns to narrow technical investigations and allocate resources efficiently.

Model Limitation

While the model provides reliable directional risk ranking, its predictive strength is moderate rather than definitive. It should guide prioritization rather than replace expert judgement. It is also constrained by historical reporting consistency and limited ability to capture complex non-linear engineering interactions, so outputs are best interpreted as early-warning indicators within a broader governance and diagnostic framework, not as standalone decision rules.

Model 2 (Classification Tree – Random Forest)

Model Performance

After the model was deployed, it is observed that the model achieved 95.5% accuracy, which is a good indicator that the model will be able to predict the component that failed based on the text description. Other metrics were measured, like balanced accuracy 93.9%, top-3 accuracy 97.0%. This prediction can reduce the costs of investigation and narrow down the path to investigate which component has a failure (Leading to the most possible component out of the 137 components in the model).

In comparison between Random Forest and the Multinomial Logistic Regression, these are the model performance comparisons.

Metric	Multinomial Logistic Regression	Random Forest	Delta
Test Accuracy	49.60%	95.50%	+45.9 pts
Balanced Accuracy	52.70%	93.90%	+41.0 pts
Top-3 Accuracy	63.80%	97.00%	+33.2 pts
Weighted AUC	0.9240	0.997	+0.073

Table 1 Comparison between Multinomial Logistic Regression and Random Forest Model

Model Interpretation and Business Insights

With the Random Forest prediction model, the complexity of analysis can understand and simplify the recall problem by giving importance scores to each variable's contribution and predict the root cause of the recall. The diversity of the number of trees applied in the model (n=300) helps to improve the complementary patterns. With a test accuracy of 95.5%, the model can boost the efficiency of the analysis team for each analysis.

The 20-day period is a time frame based on Ni e Huang (2018) studies that point that 21.7% of the recalls are analysed in a 1–2-month scope of time, so in an optimistic scenario a recall can be analyzed in 20 days. With the model, the time of the starting analysis can be reduced.

Key Executive Insight

It is possible to reduce the scope of the analysis of the recall team from 20 working days to 1.5 days, based on the following assumptions Hourly Cost: US$ 62.64/Hour (75th percentage of a Mechanical Engineer – OCC_CODE: 17-2141 (U.S. Bureau of Labor Statistics (2024)), with an estimated overhead costs of 35% (Which falls within the cost range reported by Weltman (2019), leading to a daily cost of US$ 676.51 and 20 day cost of US$13,530.24.

Cost Description	Value		Type	Percentage
Hourly Cost	$	62.64	US$	-
Overhead Cost	$	21.92	US$	35%
Hourly Cost + Overhead Total	$	84.56	US$	-
Number of Hours		8	Hours	-
Daily Cost	$	676.51	US$	-
Number of Days		20	Days	-
Cost of Analysis	$	13,530.24	US$	-

Table 2 Total Estimated Costs of Recall Analysis

After all these considerations, with the help of the model recalls analysis can take 1.5 days, reducing investigation costs to US$ 1,014.77 (92.5% reduction from the 20-day investigation estimated costs).

Cost Description	Value		Type
Daily Cost	$	676.51	US$
Number of Days		1.50	Days
Cost of Analysis with Model	$	1,014.77	US$
Cost of Analysis without Model	$	13,530.24	US$
Savings		-92.5%	Percentage

Table 3 Total Estimated Savings of Recall Analysis with the Classification Model (Random Forest)

Model Limitation

The model and the savings have some limitations that need to be tested. The estimated savings model has some limitations as well, since the actual period that the recall analysis takes to be finished is an estimation, and the hourly cost of an engineer is also an estimation. For Ford's scenario, the next step is to obtain the data of how long it takes to do an analysis and the real hourly cost to do a precise estimation of the potential savings that the model can provide. Also, the number of 1.5 days after the implementation of the model needs to be verified in a pilot project.

Analysis of Strategic Options

Model 1 (Binary Logistic Regression)

The analytical results indicate that recall escalation risk is primarily driven by recurring platform patterns and platform–component interactions rather than isolated component failures. This finding implies that Ford's quality challenges are structural and systemic rather than purely episodic. Consequently, effective organizational response should focus on platform-level governance, predictive monitoring, and targeted engineering interventions rather than reactive case-by-case investigation alone. The logistic regression model provides a probabilistic early-warning signal that can inform multiple viable strategic pathways.

Option 1 – Platform-Centric Preventive Quality Program

This option involves restructuring quality assurance processes so that recurring vehicle platforms identified by the model receive priority engineering audits, supplier reviews, and design validation cycles before new production runs. The primary advantage is that it directly addresses the strongest statistical driver of escalation and therefore offers the highest potential reduction in severe recalls. It also creates long-term structural improvement and reputational benefits. However, the disadvantage is that it requires cross-departmental coordination, longer implementation timelines, and higher upfront engineering and supplier-management costs. Expected benefits include reduced recall frequency, lower warranty liabilities, and stronger regulatory compliance positioning. Key risks include misallocation if future platform patterns shift and uncertainty due to moderate model discrimination rather than perfect prediction.

Option 2 – Predictive Risk-Triage Integration in Recall Operations

Under this approach, the logistic regression model is embedded into existing recall-management dashboards as a prioritization tool rather than a design-change driver. The main advantage is rapid implementation with relatively low cost, as it leverages existing operational structures while improving decision speed and resource allocation efficiency. It enhances early-warning visibility without major organizational restructuring. The limitation is that it treats symptoms rather than root design causes, meaning systemic platform issues may persist. Benefits include faster investigation cycles, reduced analysis costs, and improved managerial responsiveness. Risks include over-reliance on probabilistic scores and potential complacency if human oversight is reduced.

Option 3 – Component-Focused Corrective Engineering Strategy

This option centers on analyzing component categories highlighted by the model after platform risk is identified, directing technical teams to refine specific part designs or supplier contracts. Its advantage lies in technical precision and measurable short-term engineering improvements. However, the analytics show that component variables alone have weaker predictive strength, meaning this strategy may yield incremental rather than transformational gains. Costs are moderate and primarily engineering-driven, while benefits include targeted defect reduction and supplier performance improvement. The principal risk is that it underestimates systemic platform drivers and therefore fails to prevent broader escalation patterns.

Recommended Strategic Direction

The recommended course of action is a combined Platform-Centric Preventive Program supported by Predictive Risk-Triage Integration. The model evidence consistently shows that platform behavior is the dominant escalation driver, while the probabilistic scoring system provides timely prioritization signals. Integrating both approaches allows Ford to address structural root causes while simultaneously improving operational responsiveness and cost efficiency. This hybrid strategy balances long-term design governance with short-term decision agility, reducing both recall severity and investigation expenditure. Component-level analysis should remain a secondary diagnostic layer rather than the primary strategic focus.

Overall Model Limitation and Strategic Uncertainty

Although the logistic regression model provides meaningful directional insight, its discrimination power is moderate and its calibration varies across probability bands. Strategic decisions should therefore treat model outputs as risk indicators rather than deterministic forecasts. Future shifts in manufacturing practices, supplier networks, or regulatory reporting standards may alter platform risk behavior, introducing uncertainty into long-term projections. Consequently, periodic model retraining, expert validation, and complementary analytical tools are essential to sustain decision accuracy and organizational resilience.

Model 2 (Random Forest Classification)

The accuracy of 95.5% success of the model is an excellent test to predict components failure categories based on recall text descriptions, with an estimated reduction to 1.5 days and reducing the cost of analysis (US$12,515 per recall). However, these numbers are assumptions and need to be tested.

Option 1 – Validation of the Model followed by Integration

The first strategic option for the Random Forest model is the validation of the model based on Ford's Data Base, testing the assumptions of Cost Reduction (Need to validate if the Number of Days per analysis and the cost of quality engineer personal are credible estimations). After this initial validation, a pilot can be implemented with the integration of the model to the recall investigation team, focusing on a sample of 30 cases to validate the model (In the expected period of 6 months). After this pilot phase, the expansion of the model to the whole team and constant monitoring of the analysis from a third party to ensure the model is a fit for the expected purposes. Lastly, the last phase (Starting of the second year), that integrates the model with upstream quality processes, prioritizing highest-impact use cases such as supplier quality feedback for top Pareto components (sensing systems, software, airbags, transmissions) rather than attempting comprehensive enterprise transformation.

The advantage of this approach is risk mitigation – From the sequence of Ford recall events, the need for quick and precise analysis can help Ford Company to reduce the risks of bad classifications and at the same time, improve efficiency of the quality team. On the other hand, the model has some risks due to the need of validation and the limitation to 137 components (Recalls from 2015 to 2026) and might not be able to identify new technologies, so an improvement of the model should be on sight to avoid misleading information on the recall department.

Option 2 – Immediate Integration of the Model to the production line

This approach integrates the model directly into the quality recall team, ensuring that the whole team has access to the model and could gain proper knowledge of the tool. In this way, the validation of the model is skipped and risks of model accuracy on Ford Data need to be alerted to the C-Level team, since it's a risk of reputation that could lead to improper qualification due to the quick integration in the quality Team.

The benefit of this approach is a faster implementation strategy, reducing the option 1 by 6 months and getting earlier results, but at the same time, with higher risks of improper classifications and misleading information to the quality team.

Recommended Strategic Direction

Option 1 is the best cost-high quality option due to the validation, "Validation of the Model followed by Integration". The accuracy of 95.5% is a strong indicator for a excellent model that can aid the quality team to reduce the number of days that the analysis takes. This strategy shares cost efficiency with risk management towards Ford recalls issue, meanwhile building organizational capability. Immediate integration should be avoided until pilot evidence confirms investigation timeline assumptions and model generalization to Ford's documentation.

Component classification is a model to complement the platform-centric quality governance identified in Model 1, working as a diagnostic aid rather than the standalone quality management system.

Overall Model Limitation and Strategic Uncertainty

Assumptions of Cost of Engineering team, NHTSA data instead of Ford Data, Number of Days per Recall Analysis are variables that need to be validated for this model to prove its accuracy.

In the management perspective, the model should not replace teams and be considered a milestone on recall analysis. The model itself should be viewed as a crucial diagnostic tool that will make the quality team more efficient.

Recommendation and Implementation Plan

Ford should immediately implement a Hotspot-Focused Quality Risk Reduction Program that concentrates engineering, supplier quality, and governance resources on the highest-risk platform and component clusters identified in the 2015–2025 recall analysis. The evidence presented in Section "Analysis of Strategic Options" demonstrates that recall exposure is structurally concentrated rather than evenly distributed. A limited number of recurring platform and component interactions account for a disproportionate share of severity-weighted recall risk and financial impact. Because the current quality cost crisis is driven by recurrence and escalation patterns embedded within specific vehicle platforms, the most effective and defensible course of action is targeted structural intervention at those hotspots.

This recommendation is directly supported by the strategic options assessment. While the predictive early-warning option offers longer-term preventive capability and the governance reset option strengthens compliance posture, neither addresses the immediate concentration of exposure currently driving financial volatility. The hotspot strategy converts analytical insight into focused operational execution. It aligns resource allocation with measured risk intensity and therefore provides the strongest near-term impact on warranty cost stabilization and recall severity reduction.

The justification rests on three considerations. First, the financial exposure is material and sustained, exceeding one billion dollars in recent warranty and field service impacts. Second, the analysis shows that recurrence and severity, not recall count alone, are the dominant drivers of cost escalation. Third, risk concentration creates leverage. Reducing exposure within a small subset of high-impact clusters produces disproportionate financial benefits relative to required investment. The problem is systemic but not diffuse, and intervention must therefore be precise rather than broad.

Implementation should proceed in structured phases over an eighteen-month horizon. During the first three months, Ford should establish a Platform Risk Review Steering Committee embedded within existing quality governance structures and reporting to executive leadership. This body will own the Severity-Weighted Exposure Index developed in the analytical phase and generate a ranked High-Risk Platform Watchlist. The initial focus should be the top five platform and component clusters that demonstrate persistent recurrence across both the full 2015–2025 dataset and the recent 2021–2025 escalation window. Executive sponsorship at this stage is critical to ensure accountability across engineering, manufacturing, compliance, and supplier management functions.

Between months three and nine, Ford should conduct structured deep-dive audits of each prioritized hotspot cluster. These audits should evaluate design validation robustness, supplier defect recurrence, integration testing processes, and escalation pathways for early defect signals. Particular attention should be given to software and hardware interaction risks, which increasingly drive complex recall campaigns. Engineering resources should be temporarily reallocated to stress-test recurrence drivers within affected platforms. Where recurrence is linked to supplier performance, contractual review mechanisms should incorporate escalation triggers based on recurrence thresholds rather than isolated defect events. In parallel, recurrence-informed checkpoints should be embedded into product development stage gates for upcoming platform updates. This ensures that prevention becomes embedded within the forward lifecycle rather than remaining reactive.

The expected benefits significantly exceed implementation costs. If hotspot intervention reduces severity-weighted recall exposure by five percent relative to the 1.9 billion dollars excess warranty benchmark, annual savings would approximate ninety-five million dollars. At an eight percent reduction, savings could exceed one hundred and fifty million dollars annually. Even conservative improvement therefore delivers a strong return on investment. Beyond direct financial benefit, the program is expected to stabilize warranty accrual volatility, reduce dealer service bottlenecks, and improve regulator confidence through structured risk prioritization. Importantly, this recommendation does not rely on speculative predictive uplift. It acts directly on observed concentration patterns derived from the dataset.

The implementation timeline is sequenced for measurable impact. Months zero to three focus on governance alignment and hotspot identification. Months three to nine execute deep-dive audits and corrective action design. Months nine to eighteen embed recurrence-informed quality gates and supplier escalation protocols into ongoing production and refresh cycles. Early stabilization effects should begin emerging within twelve months, with clearer financial impact visible across subsequent warranty accrual cycles.

Several critical success factors will determine delivery effectiveness. Executive sponsorship is essential to overcome isolated ownership of platform risk. Cross-functional integration between engineering, supplier quality, compliance, and analytics must be formally structured. Data transparency is required to continuously update recurrence indicators and re-rank exposure clusters. Supplier accountability mechanisms must incorporate recurrence-linked performance thresholds. Governance cadence must be institutionalized to prevent the program from becoming a one-time remediation effort.

In conclusion, the Hotspot-Focused Quality Risk Reduction Program represents the most immediate, financially defensible, and operationally coherent response to Ford's quality-driven recall exposure. The analysis demonstrates that risk is concentrated, persistent, and materially linked to warranty volatility. By concentrating intervention where recurrence and severity intersect, Ford can reduce recall escalation, stabilize cost performance, and strengthen regulatory standing. Given the scale of recent financial impact and governance scrutiny, immediate implementation is justified and strategically necessary.

AI Methods Explainer

AI tools were used only as support tools in this project. They were not used to make final decisions or produce model results. The main analysis, coding, and interpretation were completed by the group. AI was used to assist with efficiency and clarity.

For coding, AI was sometimes used as a guide to check syntax in R and Python. It helped suggest possible fixes when code errors appeared. The scripts for EDA and Model 1 were written and tested in R by us. The Model 2 was written and tested in Python by us. Feature engineering steps and the data preparation workflow were first discussed and agreed by the group. AI then helped convert this workflow into correct code syntax. After that, we manually ran the code in the IDE and checked whether the outputs matched our expectations. Any AI suggestion was reviewed before use. If results looked inconsistent or illogical, the code was revised or rejected.

we compared all outputs with our own judgement and statistical logic. Visualisations, metrics, and tables were generated directly from our datasets. We verified that numbers were consistent across different runs and tools. AI did not influence the choice of variables, thresholds, or model structure.

AI was also used in the early stage to suggest possible theoretical and analytical approaches that might fit the problem. It provided several options, but the group did not accept them naively. We checked academic references, read the suggested frameworks, and then discussed together before selecting the final approach. The final decision was based on group judgement, not AI output.

AI was further used for grammar checking and minor writing refinement. It helped compress some sentences when the report exceeded the word limit. It did not generate analytical conclusions or business recommendations.

Overall, AI was used in a limited and controlled manner. It supported proofreading, formatting, and small technical guidance. It did not replace human judgement, analytical reasoning, or data preparation steps. All core analysis, modelling decisions, and validation processes remained human driven.

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Appendices

Appendix A

{"Count":1,"Message":"Results returned successfully","results":[{"Manufacturer":"BMW OF NORTH AMERICA, LLC","NHTSACampaignNumber":"12V176000","parkIt":false,"parkOutSide":false,"overTheAirUpdate":false,"ReportReceivedDate":"20/04/2012","Component":"SEATS:FRONT ASSEMBLY:HEAD RESTRAINT","PotentialNumberofUnitsAffected":7600,"Summary":"BMW IS RECALLING CERTAIN MODEL YEAR 2012 BMW 3-SERIES VEHICLES MANUFACTURED FROM OCTOBER 19, 2011, THROUGH MARCH 18, 2012, THAT HAVE FRONT SEAT HEAD RESTRAINTS THAT EXCEED THE DOWNWARD MOVEMENT LIMIT OF 25MM IN THE HIGHEST POSITION. THUS, THESE VEHICLES FAIL TO COMPLY WITH THE REQUIREMENTS OF FEDERAL MOTOR VEHICLE SAFETY STANDARD NO. 202A,\"HEAD RESTRAINTS.\"","Consequence":"IN THE EVENT OF A VEHICLE CRASH, THE HEAD RESTRAINT MAY UNEXPECTEDLY MOVE DOWN SLIGHTLY IF IT WAS ADJUSTED TO THE FULLY EXTENDED POSITION, INCREASING THE RISK OF PERSONAL INJURY.","Remedy":"BMW WILL NOTIFY OWNERS, AND DEALERS WILL ATTACH A CLAMP TO THE FRONT SEAT HEAD RESTRAINT POSTS, FREE OF CHARGE. THE SAFETY RECALL BEGAN ON MAY 30, 2012. OWNERS MAY CONTACT BMW CUSTOMER RELATIONS AND SERVICES AT 1-800-525-7417.","Notes":"CUSTOMERS MAY CONTACT THE NATIONAL HIGHWAY TRAFFIC

SAFETY ADMINISTRATION'S VEHICLE SAFETY HOTLINE AT 1-888-327-4236 (TTY: 1-800-424-9153); OR GO TO HTTP://WWW.SAFERCAR.GOV.","ModelYear":"2012","Make":"BMW","Model":"3-SERIES"}]}.

Appendix B

""" NHTSA Recall Data Fetcher ========================= This script reads campaign URLs from an Excel file, fetches recall data from the NHTSA API, and compiles all information into a comprehensive spreadsheet.

Requirements: pip install pandas openpyxl requests tqdm

Usage: python nhtsa_recall_fetcher.py """

import pandas as pd import requests import time import json from datetime import datetime from pathlib import Path from tqdm import tqdm import re

def extract_campaign_number(url: str) -> str: """Extract campaign number from NHTSA API URL.""" match = re.search(r'campaignNumber=(\w+)', url) return match.group(1) if match else None

def fetch_recall_data(campaign_number: str, max_retries: int = 3) -> dict: """ Fetch recall data from NHTSA API for a given campaign number.

```
Args:
    campaign_number: The NHTSA campaign number (e.g., '26E003000')
    max_retries: Maximum number of retry attempts

Returns:
    Dictionary containing recall data or error information
"""
url =
f"https://api.nhtsa.gov/recalls/campaignNumber?campaignNumber={campaign_number}"

for attempt in range(max_retries):
    try:
        response = requests.get(url, timeout=30)
        response.raise_for_status()
```

```
            data = response.json()
            return {
                'success': True,
                'data': data,
                'error': None
            }
        except requests.exceptions.RequestException as e:
            if attempt < max_retries - 1:
                time.sleep(2 ** attempt)  # Exponential backoff
            else:
                return {
                    'success': False,
                    'data': None,
                    'error': str(e)
                }
    return {'success': False, 'data': None, 'error': 'Max retries exceeded'}


def flatten_recall_record(record: dict, campaign_number: str) -> dict: """ Flatten a single recall record into a dictionary for DataFrame conversion.

Args:
    record: Single recall record from API response
    campaign_number: The campaign number for reference

Returns:
    Flattened dictionary with all fields
"""
# Define all possible fields from NHTSA API
fields = [
    'NHTSACampaignNumber',
    'NHTSAActionNumber',
    'ReportReceivedDate',
    'Component',
    'Summary',
```

```
    'Consequence',
    'Remedy',
    'Notes',
    'Manufacturer',
    'ModelYear',
    'Make',
    'Model',
    'ParkIt',
    'ParkOutSide',
    'NHTSAId',
    'potentialNumberofUnitsAffected'
]

flat_record = {'CampaignNumber': campaign_number}

for field in fields:
    flat_record[field] = record.get(field, '')

return flat_record


def process_api_response(response_data: dict, campaign_number: str) -> list: """ Process
API response and extract all recall records.

Args:
    response_data: Full API response
    campaign_number: Campaign number for reference

Returns:
    List of flattened records
"""
records = []

if response_data and 'results' in response_data:
    results = response_data['results']
```

```
    if isinstance(results, list):
        for result in results:
            flat_record = flatten_recall_record(result, campaign_number)
            records.append(flat_record)
    elif isinstance(results, dict):
        flat_record = flatten_recall_record(results, campaign_number)
        records.append(flat_record)

# If no records found, create a placeholder with just the campaign number
if not records:
    records.append({
        'CampaignNumber': campaign_number,
        'NHTSACampaignNumber': campaign_number,
        'Error': 'No data found'
    })

return records


def main(): # Configuration INPUT_FILE = 'MISSING_RECALLS.xlsx' OUTPUT_FILE =
f'NHTSA_Recalls_Complete_{datetime.now().strftime("%Y%m%d_%H%M%S")}.xlsx'
DELAY_BETWEEN_REQUESTS = 0.5 # seconds - adjust to avoid rate limiting

print("=" * 60)
print("NHTSA Recall Data Fetcher")
print("=" * 60)

# Check if input file exists
if not Path(INPUT_FILE).exists():
    print(f"Error: Input file '{INPUT_FILE}' not found.")
    print("Please ensure the file is in the current directory.")
    return

# Read input file
print(f"\nReading input file: {INPUT_FILE}")
```

```
df_input = pd.read_excel(INPUT_FILE)

if 'URL' not in df_input.columns:
    print("Error: 'URL' column not found in input file.")
    return

urls = df_input['URL'].tolist()
total_urls = len(urls)
print(f"Found {total_urls} URLs to process")

# Extract campaign numbers
campaigns = []
for url in urls:
    campaign = extract_campaign_number(url)
    if campaign:
        campaigns.append(campaign)
    else:
        print(f"Warning: Could not extract campaign number from URL: {url}")

print(f"Extracted {len(campaigns)} valid campaign numbers")

# Fetch data for each campaign
all_records = []
errors = []

print(f"\nFetching recall data from NHTSA API...")
print(f"(Delay between requests: {DELAY_BETWEEN_REQUESTS}s)")
print("-" * 60)

for i, campaign in enumerate(tqdm(campaigns, desc="Processing campaigns")):
    # Fetch data
    result = fetch_recall_data(campaign)

    if result['success']:
```

```
            records = process_api_response(result['data'], campaign)
            all_records.extend(records)
        else:
            errors.append({
                'CampaignNumber': campaign,
                'Error': result['error']
            })
            # Still add a record with error info
            all_records.append({
                'CampaignNumber': campaign,
                'NHTSACampaignNumber': campaign,
                'FetchError': result['error']
            })

        # Rate limiting delay
        if i < len(campaigns) - 1:
            time.sleep(DELAY_BETWEEN_REQUESTS)

print("-" * 60)
print(f"\nProcessing complete!")
print(f"  - Total records fetched: {len(all_records)}")
print(f"  - Errors encountered: {len(errors)}")

# Create DataFrame
df_output = pd.DataFrame(all_records)

# Reorder columns to prioritize important fields
priority_columns = [
    'CampaignNumber',
    'NHTSACampaignNumber',
    'Make',
    'Model',
    'ModelYear',
    'Manufacturer',
```

```
    'Component',
    'Summary',
    'Consequence',
    'Remedy',
    'Notes',
    'ReportReceivedDate',
    'potentialNumberofUnitsAffected',
    'ParkIt',
    'ParkOutSide',
    'NHTSAActionNumber',
    'NHTSAId'
]

# Get all columns and reorder
all_columns = df_output.columns.tolist()
ordered_columns = [col for col in priority_columns if col in all_columns]
remaining_columns = [col for col in all_columns if col not in ordered_columns]
final_columns = ordered_columns + remaining_columns

df_output = df_output[final_columns]

# Save to Excel with formatting
print(f"\nSaving to: {OUTPUT_FILE}")

with pd.ExcelWriter(OUTPUT_FILE, engine='openpyxl') as writer:
    # Main data sheet
    df_output.to_excel(writer, sheet_name='Recall Data', index=False)

    # Errors sheet (if any)
    if errors:
        df_errors = pd.DataFrame(errors)
        df_errors.to_excel(writer, sheet_name='Errors', index=False)

    # Summary sheet
```

```
    summary_data = {
        'Metric': [
            'Total URLs Processed',
            'Total Records Retrieved',
            'Unique Campaigns',
            'Unique Makes',
            'Unique Models',
            'Errors',
            'Processing Date'
        ],
        'Value': [
            total_urls,
            len(df_output),
            df_output['CampaignNumber'].nunique() if 'CampaignNumber' in df_output.columns
else 0,
            df_output['Make'].nunique() if 'Make' in df_output.columns else 0,
            df_output['Model'].nunique() if 'Model' in df_output.columns else 0,
            len(errors),
            datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        ]
    }
    df_summary = pd.DataFrame(summary_data)
    df_summary.to_excel(writer, sheet_name='Summary', index=False)

print(f"\n{'=' * 60}")
print("SUCCESS! Output file created.")
print(f"{'=' * 60}")

# Display sample of data
print("\nSample of retrieved data:")
print("-" * 60)
if 'Make' in df_output.columns and 'Model' in df_output.columns:
    sample_cols = ['CampaignNumber', 'Make', 'Model', 'ModelYear', 'Component']
    available_cols = [c for c in sample_cols if c in df_output.columns]
```

```
    print(df_output[available_cols].head(10).to_string(index=False))
else:
    print(df_output.head(10).to_string(index=False))


if name == "main": main()
```